

Health, Inequality, and Healthy Ageing in Italy

Xiaotian Tian

1 Introduction and Literature

Italy is one of the world’s oldest societies. In 2025 its median age is 49.1, the highest in Europe, and its old-age dependency ratio is 39% (European Commission, 2026; Istat, 2026). Population ageing makes the length of working life a central fiscal and distributional question. It also raises a harder empirical question: when older workers leave employment, how much of that exit reflects deteriorating health, and how much reflects pension rules, socio-economic resources, and labour-market opportunity costs?

Italy is a useful setting for this question. It combines steep socio-economic health gradients, historically low female labour-force participation, heavy reliance on family care, and a large pension shock. The *Fornero reform* (Law 214/2011) raised statutory retirement ages, tightened early-exit routes, and abruptly delayed pension access for many older workers (Boeri & Brugiavini, 2008; Brugiavini et al., 2025). These features allow the paper to study health inequality and retirement institutions in the same empirical setting.

The analysis connects three literatures. Lifecycle labour-supply models treat health as a key state variable for work and retirement decisions (French, 2005). Pension-incentive studies show that retirement ages and institutional rules can strongly shift older-age employment, including in Italy (Brugiavini et al., 2019, 2021). The biological-age and frailty literatures argue that multidimensional health can be summarised in indices that predict functional decline better than chronological age alone (Rockwood & Mitnitski, 2007; Schindler, 2025; Scott et al., 2021). The health-inequality literature shows that socio-economic gradients in morbidity and frailty open early and persist over the lifecycle (Wagstaff et al., 1991). This paper brings these literatures together by asking whether measured health inequality is large enough, and binding enough, to explain later-life labour supply.

The paper is organised as follows. Section 2 describes age profiles and inequality gradients in Italian health using easySHARE waves 1–9. The following Section 3 studies two-year mobility onset, first with nested logits and a discrete-time hazard model and then with machine-learning benchmarks. We then construct a PCA-based health index in Section 4, compare it with a Rockwood deficit-accumulation frailty index, and use the composite measure to document health inequality across gender, education, income, and urbanicity in Section 5. The final empirical Section 6 links health to economic outcomes: labour-force participation, weekly hours, health-shock responses, retire-by-65 predictions, counterfactual health-equalisation exercises, and the Fornero reform.

The results point to a distinction between health as an inequality outcome and health as a retirement constraint. Health gradients by gender, education, and income are large, and health variables strongly predict mobility onset. Health also predicts labour-force participation in pooled data. Yet the estimated labour-supply response to within-person health changes is small, health shocks have limited average effects, and equalising low- and medium-educated respondents’ health to the high-educated benchmark does not produce a sizeable delay in retirement by age 65. By contrast, statutory eligibility and the Fornero reform are strongly associated with labour-force participation near retirement ages. The evidence therefore suggests that, in Italy, health inequality is central to welfare and healthy ageing, but pension institutions and socio-economic resources are central to the retirement margin.

2 Describing Italian Ageing and Health (Step 1)

To begin our analysis, we pool samples across waves and explore how health indicators change as people age. As shown in Figure 1, health quickly worsens in the ageing process. For most of the indicators, women suffer from

faster deteriorating trends as they become older. Women also have more mobility limitations and higher degrees of depression compared to men, across all age bins. Although they seem to enjoy fewer chronic conditions and better abilities to recall initially, this reverses as they get older. Those patterns are consistent with lower reported health from women.

Large socio-economic gradients are also observed. In Figure 2, we document ageing profiles by persistent income quartile, constructed from each respondent's average income decile across observed waves. Individuals from higher levels in the income hierarchy enjoy better health in all aspects. This echoes evidence that health gradients by economic status emerge early and widen over the lifecycle (Case & Deaton, 2005; Case et al., 2002). Yet there is no obvious slowdown of ageing for high income earners. Using the level of education instead of income yields similar outcomes (Figure A.1 in appendices).

On the time dimension, Figure A.2 in appendices compares the age profiles of Italy across different waves of survey. Generally, a modest healthy-ageing dividend is observed: most of the health indicators continuously improve from 2004-06 to 2019-2022. Recent cohorts enjoy fewer mobility limitation, lower degree of depression, fewer chronic conditions, and better ability to recall. This improvement is mainly characterised by shifts of the curves, and there is no obvious change in slopes, indicating a general health level rising instead of slower ageing. Self-reported health perception also improves.

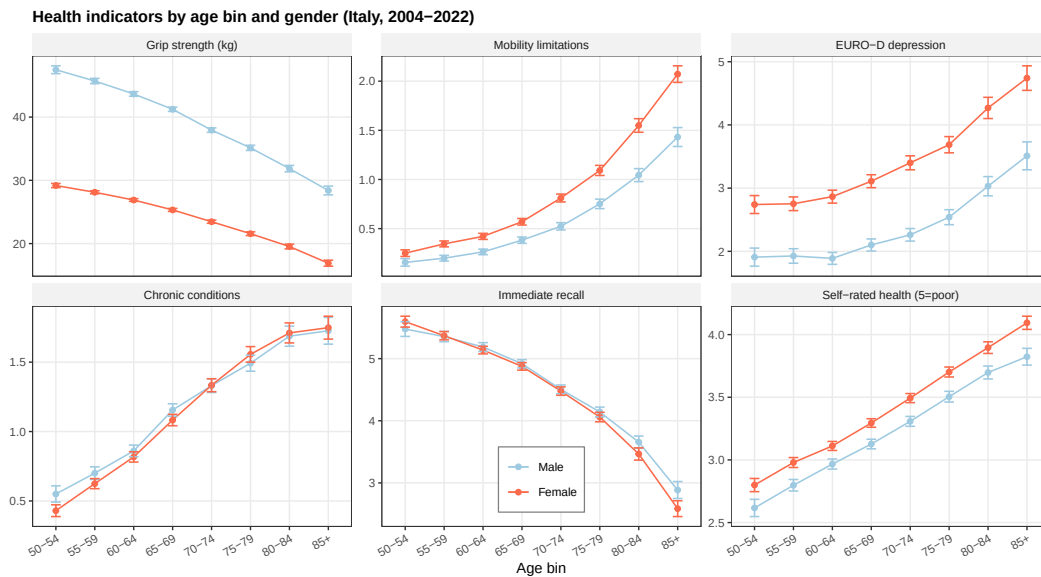


Figure 1: Age Profiles of Italy (Pooled). Summary statistics by gender reported in Table A.3.

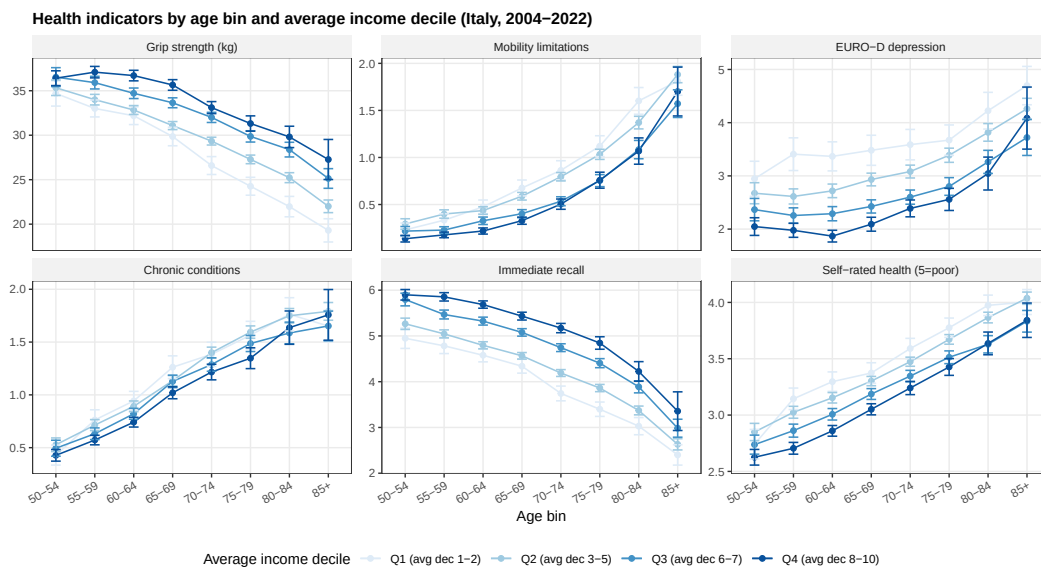


Figure 2: Age profiles of Italy by persistent income quartile. Summary statistics by wave reported in Table A.5.

3 Predicting Two-Year Mobility Onset (Step 2 & 3)

3.1 Pooled logits and a discrete-time hazard model (Step 2)

The first empirical outcome is the transition into substantial mobility limitation, i.e. *2-year mobility onset*, defined as the transition from fewer than two mobility difficulties to two or more mobility difficulties by the next SHARE wave. This is stricter than the common binary “any mobility limitation” definition and is intended to capture the onset of multiple, or substantial, mobility limitation, while keeping the event common enough for predictive evaluation¹.

We begin with applying logit models to pooled data. Predictors are added sequentially to evaluate the predictive information that they carry. As shown in Figure 3a and Table B.1, demographics alone explain a modest share of onset risk, and adding socio-economic variables and lagged health measures further improves model performance. The fully adjusted logit reaches a pseudo- R^2 of 0.132 and an AUC of 0.762 on a respondent-level 70/30 held-out test². While all lagged health measures are statistically significant predictors, higher education turns out to be the strongest predictor among socio-economic variables; the income coefficients weaken substantially once lagged health is included.

The pooled logit treats every person-wave observation as an isolated objective of prediction and can therefore count multiple at-risk spells for the same respondent if mobility recovers between waves. Since the first entrance into mobility limitation is the clinically cleaner event, we complement the logit with a *discrete-time complementary log-log* (cloglog) hazard model in the Prentice–Gloeckler form (Allison, 1982; Prentice & Gloeckler, 1978). Each respondent contributes one person-period row per wave while still at risk ($\text{mobilityind} < 2$); the row at which onset occurs carries $y_{it} = 1$, and the person exits the risk set thereafter. Writing h_{it} for the one-wave hazard, the model is

$$\log(-\log(1 - h_{it})) = f(\text{age}_{it}) + X_{it}\beta,$$

Underlying this model is a continuous-time proportional hazard model, and $\exp(\beta_k)$ reads as a hazard ratio. Age enters non-parametrically as a four-degree-of-freedom B -spline $f(\text{age}_{it})$ so that the accelerating risk above 65 shown by Figure 3b is estimated, not imposed.

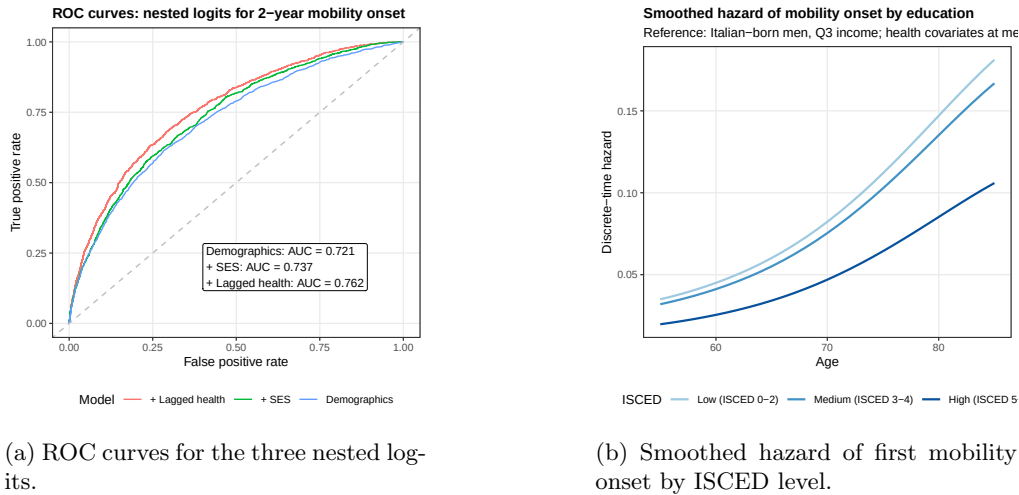


Figure 3: Predicting two-year mobility onset: nested logits and discrete-time hazard profile.

The estimates in Table B.1 are consistent across the logit and cloglog hazard specifications. Prior health dominates in terms of predictability: each additional chronic condition raises the onset hazard by 18%³, a one-point rise in

¹The choice follows the disablement process literature, where mobility limitations are treated as functional limitations preceding ADL disability (Heiland et al., 2016; Verbrugge & Jette, 1994), and is close to SHARE work that distinguishes multiple mobility limitations from milder limitation states (Litwin & Levinson, 2018; Litwin et al., 2018).

²Sample is split at the respondent level to prevent potential leakage in an observation-level split (person-wave observations from the same individual appearing on both sides of a validation boundary).

³ $\exp(0.163) - 1 \approx 18\%$. Other percentages for the cloglog model are calculated in the same way.

the EURO-D depression score raises the hazard by 10%, and each extra kilogram of grip strength lowers it by 3%; BMI and immediate recall enter with the expected signs and are statistically significant.

An education gradient is also observed. Tertiary education remains protective: high-ISCED respondents face a hazard roughly 44% lower than the low-education reference group, and this effect is highly significant across different specifications. Such inequality can be clearly seen in the smooth hazard plotted in Figure 3b. On the other hand, the income gradient weakens once lagged health is included, and the gender and migration coefficients are small and imprecise.

Overall, the models point to a clear empirical ordering: mobility onset rises steeply with age and prior health deficits, while education remains a substantial independent gradient after conditioning on measured health; income, gender, and migration add little once these margins are accounted for.

3.2 Machine-learning benchmarks (Step 3)

We use two machine-learning benchmarks to explore whether more flexible models improve predictability of mobility onset. The random forest is an ensemble of many decision trees, while XGBoost builds trees sequentially to correct earlier prediction errors (i.e. gradient boosting). Both methods can capture interactions and non-linear threshold effects without imposing them in advance. Figure 4 and Table B.2 summarise their performance⁴.

Our tree-based machine-learning models perform similarly to the logit: their AUC, Brier scores, and calibration slopes are close, with the full-scale logit taking a slight lead. This suggests that the extra flexibility of the tree models has little marginal value in this application. Similar to our logit and hazard model, the main predictive information comes from variables with a direct economic and health interpretation: age, grip strength, and the number of chronic conditions (Figure 4c). Once the logit conditions on these gradients, allowing for further flexibility and non-linearities does not materially improve out-of-sample prediction. The simpler logit is therefore the more useful benchmark: it delivers comparable predictive accuracy while maintaining interpretability.

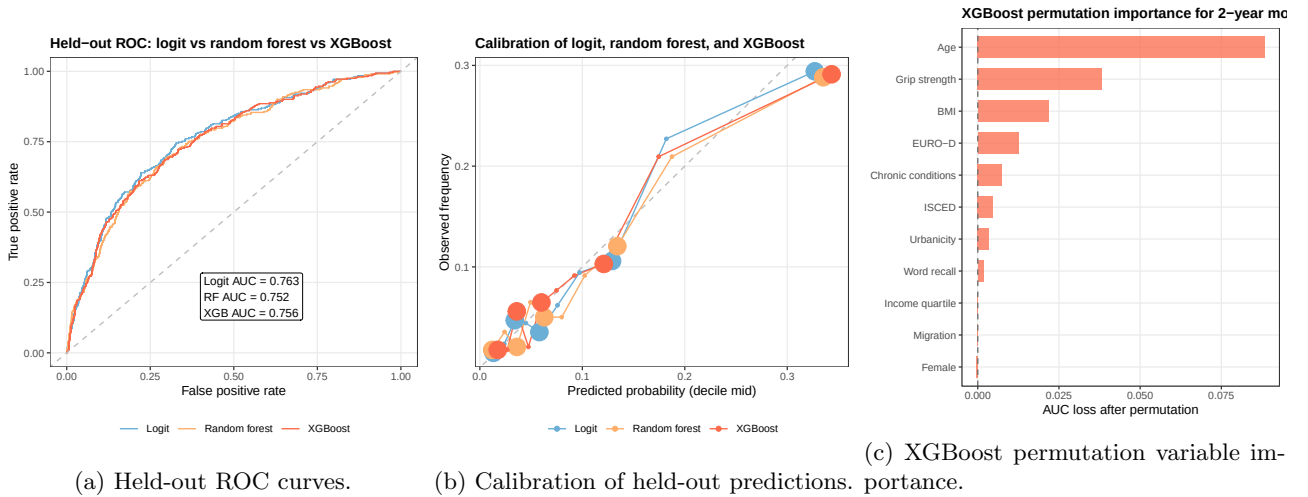


Figure 4: Held-out predictive performance for two-year mobility onset.

4 A Composite Health Index (Step 4)

In the previous section, we demonstrate that health variables carry substantial predictive information. Yet, there could be high collinearity between them, and individual indicators can be manifestations of the underlying health status (Rockwood & Mitnitski, 2007). Here, we construct a scalar health index from 12 standardised indicators using principal component analysis (PCA)⁵. We take the first principal component (PC1) as our health index: it

⁴Models use the same respondent-level train/test split as the logit. The random forest uses 1,000 trees, with the number of candidate predictors considered at each split chosen by cross-validation; XGBoost uses depth 4, learning rate 0.1, subsample and column-sample rates 0.8, with boosting rounds chosen by cross-validation.

⁵Factors included: maximum grip, immediate and delayed recall, chronic-condition count, ADL, IADL, EURO-D, mobility-limitation count, BMI, large-muscle limitations, gross motor, and fine motor. Appendix F.1 explains the selection.

captures 37% of total variance and is oriented so that higher values mean better health. Its loadings have the expected interpretation: positive on cognition and grip, negative on chronic conditions, ADL/IADL limitations, mobility limitations, and depression (see Table C.1 for specific loadings). It also shows a clear accelerating decline as age increases (Figure 5c).

Two validation checks support using PC1 as the paper’s baseline health index. First, PC1 is strongly correlated with the independently constructed 30-item Rockwood et al. (2005) frailty index ($\rho = -0.82$), so the continuous PCA and deficit-accumulation approaches recover nearly the same ordering of people up to sign (Figure 5b). Besides, replacing the nine raw health predictors in the mobility-onset logit with PC1 does not harm model performance: AUC is 0.773 with the raw health variables, 0.775 with PC1 alone, and 0.777 with Rockwood frailty (Table C.2). The scalar index therefore captures the predictive content of a set of individual health measures while keeping our models in following sections interpretable.

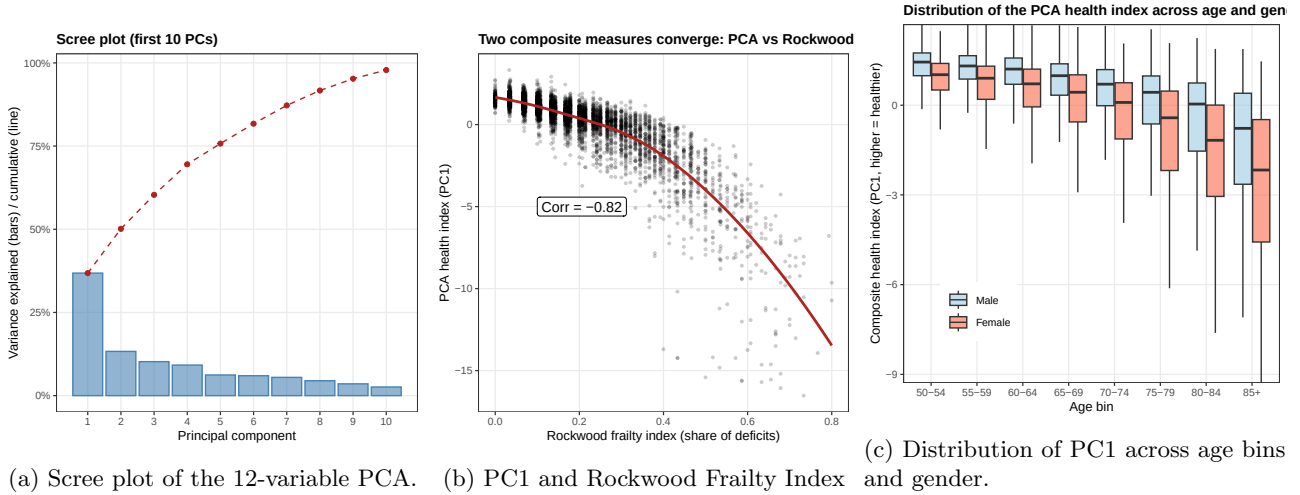
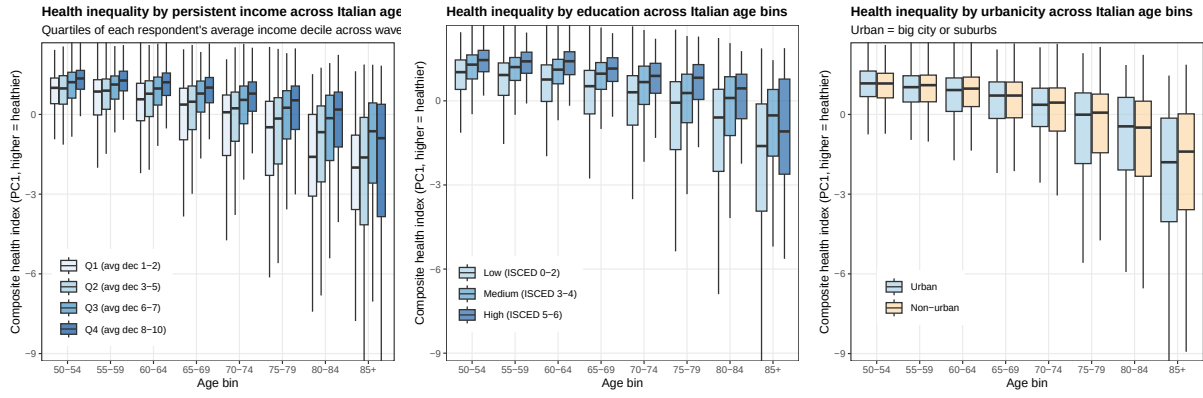


Figure 5: PCA-based composite health index and Rockwood validation.

5 Documenting Health Inequality (Step 5)

With our health index constructed, we move on to document inequalities in its distribution. The descriptive axes are gender (Figure 5c), persistent income, measured as quartiles of each respondent’s average income decile (Figure 6a); education, measured by ISCED (Figure 6b); and urbanicity (Figure 6c). Clear gender and socio-economic gradients appear in our visualisations. Gaps are already visible in the 50s and widen substantially at older ages. Men, higher-income respondents, and better-educated respondents have better initial health and slower deterioration. The urban-rural gap is less pronounced, with rural residents showing slightly better health in most age bins.

Through a different lens, the same income-health gradient is also reflected by concentration indices. Figure 7 plots the Wagstaff index and the Erreygers corrected index for Rockwood frailty by household-income decile (Erreygers, 2009; Wagstaff et al., 1991). Here, the Rockwood frailty is used rather than PC1 as the concentration indices are designed for a non-negative bounded health deficit measure. Both the Wagstaff index and the Erreygers index are negative in every wave, meaning frailty is concentrated among lower-income respondents (Appendix F.4 gives a short explanation of the concentration-index measures.). The income gradient damps between the early 2010s and 2020, then partially reappears in the 2022 wave.



(a) Health PC1 by persistent income quartile across age bins. (b) Health PC1 by education across age bins. (c) Health PC1 by urbanicity across age bins.

Note: Health PC1 by gender across age bins is reported earlier in Figure 5c. Persistent incomes are based on each respondent's average income decile across observed waves. Urban indicates big cities or suburbs.

Figure 6: Health inequality across income, education, and urbanicity.

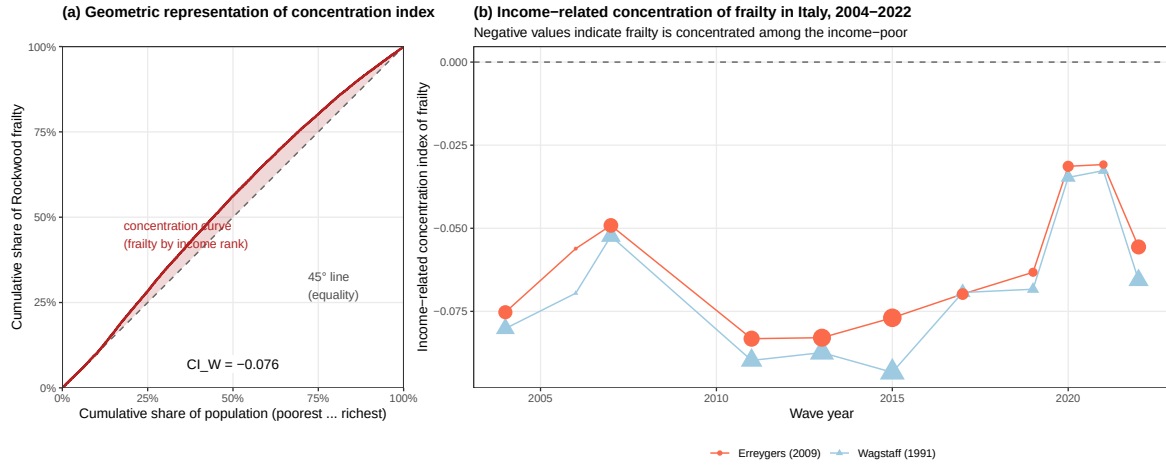


Figure 7: Income-related concentration indices of Rockwood frailty in Italy.

6 Health and Economic Outcomes (Step 6)

The last empirical section links the health gradient to labour-market outcomes. Our analysis separates three margins that are often conflated. First, we examine whether health and SES predict labour-force participation and weekly hours. Then, we explore whether discrete health shocks affect labour supply. Finally, we analyse whether equalising health at ages 50–55 would materially change later retirement and compare its effect with an institutional shock.

6.1 Health and labour force decisions

We use our health index (PC1) and Rockwood frailty alongside age, gender, education, income quartile, and migration to predict both the extensive margin and intensive margin of labour supply, i.e. labour-force participation (LFP) and weekly hours worked, at horizons $h \in \{0, 1, 2, 3\}$ SHARE waves ahead⁶. This is a *between-person* prediction exercise, aiming to predict respondents' labour-market outcomes given their current health and socio-economic profiles. As a complement, Appendix E.2 reports a *within-person* fixed-effects specification that asks whether changes in health within the same respondent move labour supply after absorbing stable person-level differences.

⁶The Italian SHARE waves are spaced approximately 2 years apart, so $h \in \{0, 1, 2, 3\}$ waves correspond to roughly 0, 2, 4, and 6 calendar years.

Here, for LFP, we estimate a pooled logit, a random forest, and XGBoost; for hours conditional on employment we estimate pooled OLS and the same tree-based models (random forest and XGBoost). Models are calibrated and evaluated using a 70/30 training/validation split at the respondent level. Figure 8 compares their performance graphically⁷.

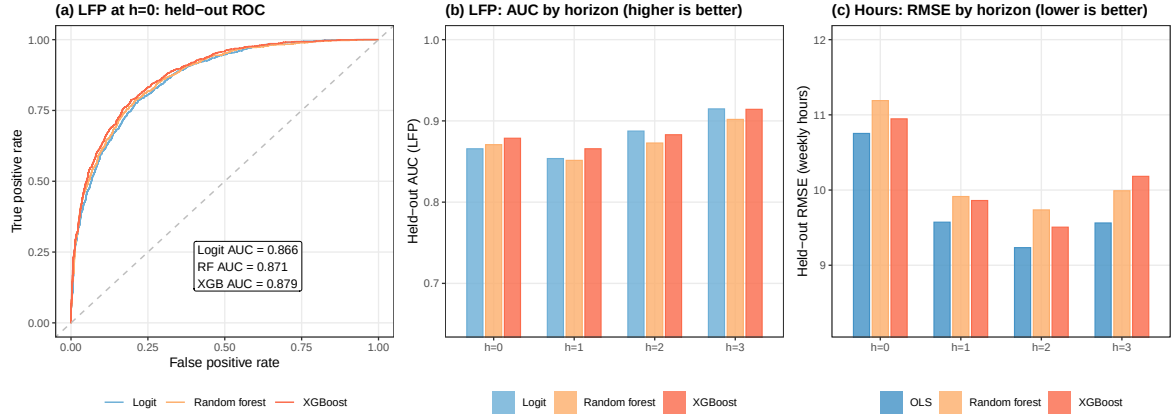


Figure 8: Held-out predictive performance of pooled logit/OLS, random forest, and XGBoost. Panel (a): ROC for LFP at $h = 0$. Panel (b): AUC for LFP, by horizon. Panel (c): RMSE for weekly hours conditional on employment, by horizon.

On model performance, the extensive margin (LFP) appears to be highly predictable: AUC ranges from 0.85 to 0.92 across horizons. XGBoost has the best performance in all benchmarks at $h = 0$ and $h = 1$, but the pooled logit catches up by $h = 2$ and is marginally best by $h = 3$. The random forest is close at the shortest horizon and then slightly below the logit. Similar to Section 3.2, a linear setup with age, gender, education, and baseline health absorbs most of the observable LFP variation, so flexible tree-based models provide only marginal improvement once the main demographic structure is included. Regarding the intensive margin (weekly hours worked conditional on employment), pooled OLS dominates the comparison by having the lowest RMSE and highest R^2 at every horizon. The random forest consistently shows the worst performance, and XGBoost only partly closes the gap. Thus, traditional econometric methods (logit and OLS) match the predictive accuracy of the machine-learning tools used here, while keeping the coefficients interpretable.

As shown in Table E.3, health contains predictive information mainly at short horizons: Health PC1 is significant for LFP at $h = 0$ and $h = 1$, while the hours coefficients are less stable. The within-person fixed-effects results in Appendix E.2 point in the same direction: after person-level differences are absorbed, the estimated association between changes in health and changes in labour supply is small, with no consistent pattern across outcomes or horizons. Overall, the implication is that the strong health–work association in later life is mostly a short-term between-person phenomenon.

6.2 Heterogeneity in the labour-supply response to health shocks

To delve deeper into the effects of health shocks on labour market outcomes and to pick up potential non-linearities in responses, we estimate an LP-DiD specification (Dube et al., 2025) for the change in labour outcome $Y \in \{\text{LFP, hours worked}\}$ in the k -th wave relative to $t - 1$ where each person’s first health shock (defined as deterioration in morbidity, physical functioning, or depressive symptoms⁸) is observed:

$$Y_{i,t+k} - Y_{i,t-1} = \beta_k \cdot \text{Treated}_{i,t,k} + \gamma_1 \text{age} + \gamma_2 \text{age}^2 + \delta_t + \varepsilon_{i,t},$$

where the treatment is experiencing the health shock. We run the same regression prior to the shock at $k = -3$ and $k = -2$ as a placebo test for the parallel trend assumption. As Figure 9 shows, this identifying assumption

⁷Numerical results can be found in appendices: see Table E.1 and Table E.2 for model performance; see Table E.3 for estimated coefficients for the pooled logit/OLS. The random forest uses 500 trees and considers three candidate predictors at each split; XGBoost uses depth 4, learning rate 0.1, and subsample and column-sample rates 0.8.

⁸See Appendix E.3 for the definition and robustness checks using alternative health shocks.

is not rejected when using LFP as the outcome variable. The post-shock estimates demonstrate only a marginal short-term effect at $k = 0$ that quickly diminishes, suggesting a small immediate LFP drop and no persistent average decline. On the intensive margin effects reported in Table E.6, effects of health shocks are even weaker with no significant coefficients for the pooled sample. These results are consistent with our conclusion in the previous section. For the average respondent, health shocks appear to play a limited role in labour-force decisions.

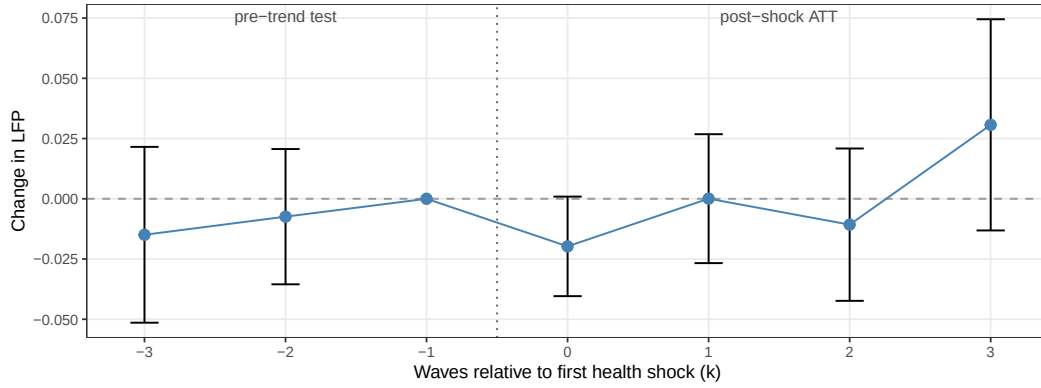
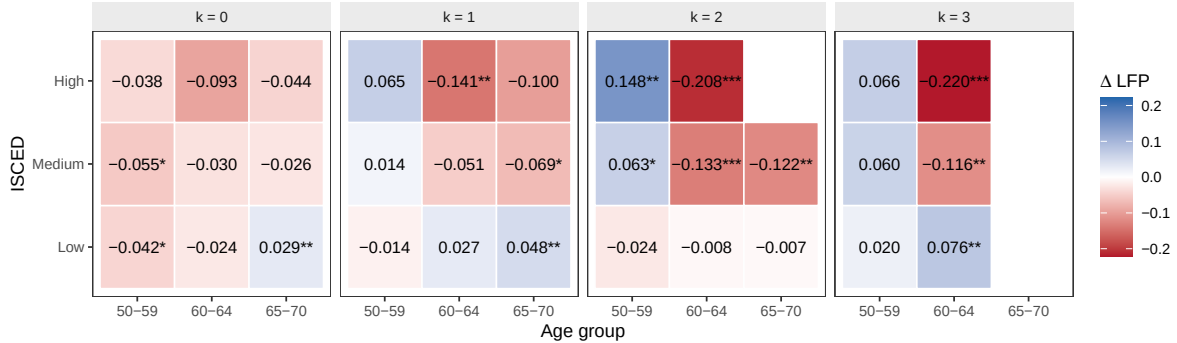


Figure 9: LP-DiD event study: change in LFP at k waves after the first health shock, Italy 50–70.

We move on to explore potential heterogeneity in the response to health shocks with a fully interacted specification accounting for different age and education groups. Figure 10 visualises the highly heterogeneous effect on LFP: for older people with high education levels, health shocks facilitate withdrawal from the labour force. A similar effect is observed for individuals above 60 with medium education levels as well. However, for people with low education levels, the incentive is much weaker, and the coefficients are even significantly positive for some subgroups. Though these results are explorative⁹, they shed light on the high educational gradient in labour supply response to health shocks in Italy, which cannot be seen using the previous specification¹⁰.



Note: Each tile is one (age group $\times$ ISCED) cell at $k = 0, 1, 2, 3$ waves after the first observed health shock; cell colour is the estimated Δ LFP and the overlaid number is the point estimate (stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$). Cells with $N < 30$ are omitted from the heatmap. Sample: Italy, ages 50–70. See Table E.6 for full numerical results.

Figure 10: LP-DiD ATT by age and education.

6.3 Health at 50–55 and later-life retirement

Finally, we investigate whether health in the early 50s explains later retirement. Because retirement can itself affect health and cognition (Coe & Zamarro, 2011; Mazzonna & Peracchi, 2012), we use health measured before the retirement window rather than contemporaneous post-retirement health. We take each person's first observation at ages 50–55 and compare logit, random forest, and XGBoost models on their performance in predicting retirement by age 65. All three models use age, age squared, gender, ISCED, income quartile, migration, Health PC1, and

⁹Rigorously speaking, the placebo test for parallel pre-trends fails for some subgroups (see Table E.6), so we classify this heatmap as explorative rather than definitive.

¹⁰In Appendix E.4, we also try the income-age stratification, and the income gradient turns out to be noisier than the education gradient.

a statutory-age-at-65 dummy¹¹ as features, and are evaluated using the same respondent-level train/test split¹². The discrimination and calibration are shown in Figure 11, and numerical results and coefficients can be found in appendices (Table E.9 and Table E.10).

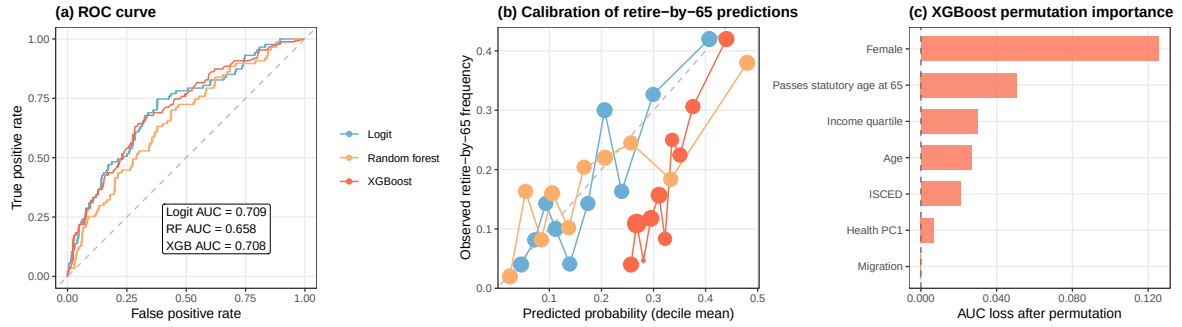
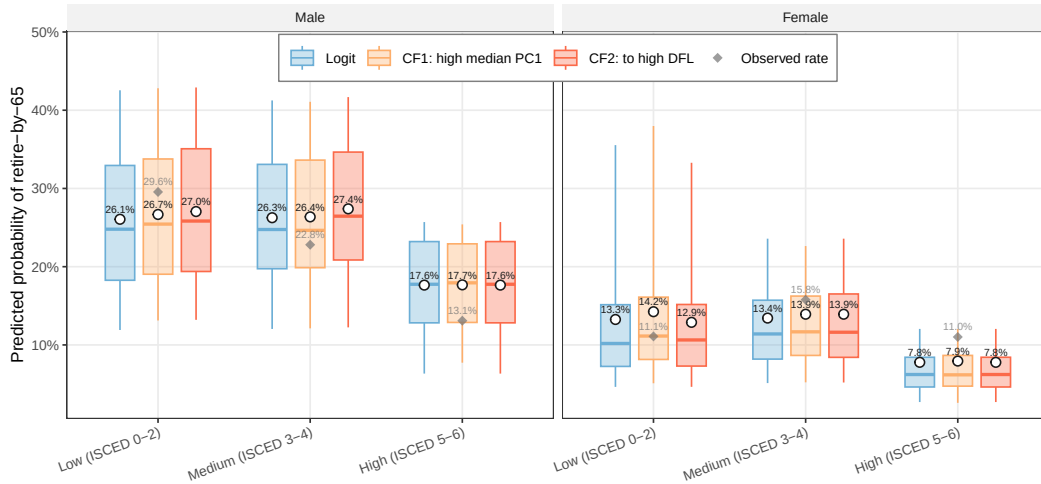


Figure 11: Retire-by-65 prediction: model performance.

The logit is the best overall predictor with the highest AUC, lowest Brier score, and a calibration slope close to one. XGBoost has nearly the same AUC but weaker calibration and a higher Brier score, while the random forest is clearly behind. In addition, logit offers interpretability of its coefficients as log odds. We therefore mainly use the calibrated logit for the counterfactuals¹³.



Note: CF1: high-ISCED median health; CF2: DFL reweighted. Each box spans the 25th–75th percentiles, the line inside the box is the median, and the whiskers span the 5th–95th percentiles; white circles show means; black diamonds show observed rates. Numerical results in Table E.11.

Figure 12: Distribution of predicted and counterfactual retire-by-65 probabilities by ISCED and gender.

To simulate the effect of health equalisation, we consider two counterfactuals. The first counterfactual (CF1) assigns everyone the high-ISCED median Health PC1. The second (CF2) uses DiNardo et al. (1996) (DFL) reweighting to give the low- and medium-ISCED groups the full high-ISCED health distribution, conditional on age and gender. A more detailed introduction of the two methods is contained in Appendix F.2. As shown in Figure 12, both counterfactuals leave the predicted retirement probability distribution almost unchanged, which is consistent with the small and statistically insignificant Health PC1 coefficient in Table E.10. In the logit counterfactuals, CF1 moves predicted retire-by-65 by less than one percentage point in every gender $\times$ ISCED cell. CF2 records slightly more heterogeneity: predicted retirement decreases only for low-educated women (13.3% to 12.9%) and increases for the other affected groups. These movements are small, but they cut against a simple health-constraint story in

¹¹The dummy equals one if a respondent aged exactly 65 would have reached or passed the statutory old-age retirement age under the actual gender/year-specific law in the calendar year when they turn 65. This schedule captures the 2011 Fornero reform and its transition.

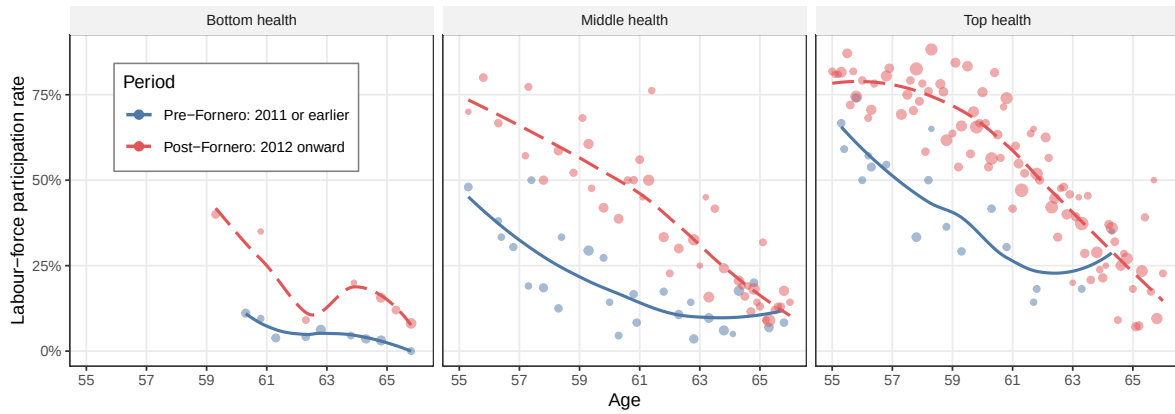
¹²The random forest uses 1,000 trees and considers three candidate predictors at each split; XGBoost uses depth 4, learning rate 0.1, and subsample and column-sample rates 0.8.

¹³We also use the XGBoost alternative in Appendix E.5.

which better health should mechanically delay retirement. Instead, the evidence is more consistent with retirement by 65 also reflecting resources and institutional access. XGBoost counterfactuals in Appendix E.5 lead to the same qualitative conclusion.

6.4 Extension: the 2011 Fornero reform as a policy shock

From the policy side, retirement appears much more responsive to pension rules than to baseline health. This is consistent with Italian and cross-country evidence that pension incentives shape employment at older ages (Brugiavini et al., 2019, 2021; Carta, 2021). Compared with our health index, the statutory-age-at-65 dummy has a large and highly significant coefficient in the logit model and ranks among the most important variables in the XGBoost permutation-importance exercise. To provide a final check on this institutional interpretation, we use the 2011 Fornero reform as a policy shock: law 214/2011 sharply raised statutory retirement ages and closed most early-exit routes from 2012 onward. Figure 13 plots labour-force participation at ages 55–66 before and after the reform, split by baseline health tercile. The post-2012 upward shift is visible across the health distribution, with the clearest separation in the middle tercile. Appendix E.6 reports both the corresponding pre/post regression and an exposure-based fixed-effects specification that isolates observations whose old-law pension access was delayed by the reform.



Note: Points are age-specific cell means for age $\times$ period $\times$ baseline-health-tercile cells with at least 20 observations. Point size is proportional to the number of observations in the cell; LOESS smooths are weighted by the same cell counts. Sample: Italy, ages 55–66. Colours distinguish pre-Fornero observations from observations in 2012 onward.

Figure 13: LFP by age and baseline health tercile, pre/post-Fornero reform.

7 Conclusion

This paper studies whether health inequality explains later-life labour supply in Italy. Health is strongly graded by age, education, income, and gender, and the PCA and Rockwood indices recover similar rankings of biological ageing. Health also predicts labour-force participation in pooled data. Yet after absorbing person fixed effects, studying health shocks, and predicting retire-by-65 from age-50–55 baselines, its marginal role becomes much smaller. For Italy, retirement models are driven more by age, SES, gender, and statutory eligibility than by the health index itself.

The counterfactuals sharpen this distinction. Equalising low- and medium-ISCED health to the high-ISCED benchmark, either through the high-ISCED median Health PC1 or DFL reweighting, does not produce a sizeable delay in predicted retirement by age 65. If anything, the results suggest that healthier and more advantaged workers have greater access to earlier retirement. The Fornero reform analysis points in the same direction. Retirement in Italy is therefore not simply a biological capacity margin; it is also a policy and resources margin.

The policy implication is that health equalisation and pension design are complements, not substitutes. Improving health remains central for welfare and healthy-life expectancy in Italy, but it is unlikely by itself to extend working lives if retirement rules and opportunity costs still favour exit.

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Structure of Appendices

The appendix is organised in six sections. Appendix A documents the data construction, variable definitions, and descriptive background. Appendices B–D report additional numerical results for mobility-onset prediction, the composite health index, and health inequality. Appendix E collects the extended labour-market and retirement analyses, including within-person regressions, LP-DiD robustness checks, counterfactual retirement tables, and the Fornero reform regressions. Appendix F gives methodological details for the PCA index, retirement counterfactuals, Rockwood frailty index, and concentration indices.

Appendix A Data and Descriptive Analysis

A.1 Data and variables

The empirical data are built from easySHARE release 9.0.0 (Börsch-Supan et al., 2013; SHARE-ERIC, 2024). Notebook 01 reads the raw Stata file, keeps Italian respondents (`country == 16`) aged 50 or above, drops value-label wrappers, and converts the easySHARE negative missing-value codes ($-1, -2, -3, -4, -7, -9, -10, \dots, -16$) to NA. The resulting person-wave panel contains 31,524 observations for 8,499 individuals, covering interview years 2004–2022. Some variables are not collected in every wave, especially around the SHARELIFE wave, so all descriptive statistics and regressions are computed on the non-missing rows relevant to each outcome rather than on a mechanically balanced panel.

The core panel is saved as `data_derived/it_panel.parquet`. It contains the variables used throughout the paper:

- ◇ Education, migration, urbanicity, and income. `iscsed3` collapses ISCED-1997 into low (0–2), medium (3–4), and high (5–6). `migration` is foreign-born if `birth_country` differs from Italy’s ISO code 380. `urban_bin` groups big cities and suburbs as urban and all other places as non-urban. `inc_dec` is the wave-specific country income decile read from `income_pct.w1–income_pct.w9`; `inc_q` groups these deciles as 1–2, 3–5, 6–7, and 8–10. For descriptive age profiles and inequality plots, we also construct a person-level average income decile and use the same four groups to proxy persistent income position.
- ◇ Demography and panel timing. `female` is recoded to a 0/1 indicator, `cohort` is `int_year - age`, and `post_fornero` equals one from 2012 onward. For each respondent we store the first observed wave and age, together with `wave_gap` and `age_gap`, so later notebooks can distinguish adjacent observations from wider calendar gaps.
- ◇ Labour-market status. `retired`, `employed`, `unemployed`, `disabled`, and `homemaker` are derived from `ep005_`. Labour-force participation is `lfp = 1` for respondents who are employed or unemployed, and weekly hours are `ep013_mod` for employed respondents only.
- ◇ Health measures. The raw health variables include self-rated health, chronic-condition count, mobility limitations, ADL/IADL counts, EURO-D depression score, grip strength, BMI, immediate and delayed recall, and three functional-limitation indices. `health_shock` marks within-person deterioration since the previous wave: at least one additional chronic condition, at least two additional mobility limitations, or crossing the EURO-D threshold from below 4 to 4 or above.

Two composite health measures are then attached. The Rockwood index, built in the setup notebook, is the mean of 30 binary deficits and is observed when at least 20 deficits are non-missing. Notebook 05 constructs `health_pc1` using complete cases on 12 standardised indicators: grip, immediate recall, delayed recall, chronic conditions, ADL, IADL, EURO-D, mobility limitations, BMI, large-muscle limitations, gross-motor limitations, and fine-motor limitations. The sign is oriented so that higher PC1 means better health. The enriched panel is saved as `data_derived/it_panel.v2.parquet`; 21,751 person-wave observations have a non-missing PC1.

The analysis samples are deliberately outcome-specific. The mobility-onset construction starts from 15,352 at-risk person-waves for 5,602 respondents: the respondent has fewer than two mobility limitations at wave t and a non-

missing next-wave mobility measure; onset is one if the next wave has two or more limitations. Complete-case model samples are smaller because the logits and hazard model require non-missing covariates. The labour-outcome notebooks keep respondents aged 50–70 and construct `lfp` and hours leads for horizons $h = 0, 1, 2, 3$ waves ahead; the machine-learning comparisons split respondents, not pooled person-waves, into train and test samples. The retire-by-65 exercise is person-level: it takes each respondent’s first age-50–55 observation, links it to whether retirement is observed by age 65, and adds a statutory-age-at-65 dummy based on the actual gender- and year-specific Italian old-age schedule in the year the respondent turns 65. That baseline sample contains 1,642 respondents.

Table A.1 lists the main constructed variables and gives plain-language definitions. Table A.2 shows the unbalanced panel structure. Table A.3 reports the person-level gender composition of the education, migration, and urbanicity groups. Tables A.4–A.5 report the pooled descriptive statistics by gender and by wave. The figures below add the descriptive age profiles used as background for the main analysis.

Name	Source	Definition
<code>mergeid</code>	easySHARE	Person identifier. This is the panel key used to link a respondent across waves and to split train/test samples at the person level.
<code>wave, int_year</code>	easySHARE	Survey wave and interview year. Italy is observed in waves 1–9, with some waves spanning two calendar years.
<code>age, female, cohort</code>	easySHARE / derived	<code>age</code> is age in years at interview. <code>female</code> is coded 1 for women and 0 for men. <code>cohort</code> = <code>int_year</code> - <code>age</code> .
<code>baseline_wave, baseline_age</code>	derived	First observed wave and first observed age for each respondent. These are used when an analysis needs person-level baseline characteristics.
<code>isced3</code>	derived from <code>isced1997_r</code>	Education grouped into Low (ISCED 0–2), Medium (ISCED 3–4), and High (ISCED 5–6).
<code>inc_dec, inc_q, avg_income_q</code>	derived from <code>income_pct_w1–income_pct_w9</code>	<code>inc_dec</code> is easySHARE’s country-wave household income decile. <code>inc_q</code> groups the current-wave decile into 1–2, 3–5, 6–7, and 8–10. <code>avg_income_q</code> applies the same grouping to each respondent’s average income decile across observed waves.
<code>migration, foreign_born</code>	derived from <code>birth_country</code>	<code>foreign_born</code> = 1 if the respondent’s birth country is not Italy (ISO code 380). <code>migration</code> labels the same distinction as Italian-born versus foreign-born.
<code>urban5, urban_bin</code>	derived from <code>iv009_mod</code>	<code>urban5</code> keeps the five original settlement-size groups. <code>urban_bin</code> groups big cities and suburbs as urban, and large towns, small towns, and rural areas as non-urban.
<code>sphus, chronic_mod, mobilityind</code>	easySHARE	Self-rated health, count of chronic conditions, and mobility-limitation count. Higher values indicate worse health for all three measures.
<code>adla, iadla, eurod</code>	easySHARE	Counts of ADL limitations, IADL limitations, and depressive symptoms on the EURO-D scale. Higher values indicate more limitations or symptoms.
<code>maxgrip, bmi</code>	easySHARE	Maximum grip strength in kilograms and body-mass index. Grip is oriented so higher values indicate better physical strength; BMI is used as a risk input rather than as a monotone health scale.
<code>recall_1, recall_2</code>	easySHARE	Immediate and delayed word-recall scores. Higher values indicate better cognitive performance.
<code>lgmuscle, grossmotor, finemotor</code>	easySHARE	Functional-limitation indices for large-muscle, gross-motor, and fine-motor tasks. Higher values indicate more functional difficulty.
<code>frailty_rockwood</code>	derived	Rockwood-style deficit index: the respondent’s mean across 30 binary health deficits, computed when at least 20 deficits are non-missing. Higher values mean greater frailty.
<code>health_pc1</code>	derived by PCA	First principal component of 12 standardised health indicators. Its sign is oriented so that higher values mean better composite health.
<code>health_shock</code>	derived	Within-person deterioration since the previous observed wave: at least one additional chronic condition, at least two additional mobility limitations, or EURO-D crossing from below 4 to 4 or above.
<code>mobility_onset</code>	derived	Outcome for the two-year mobility analysis. It equals 1 when a respondent has fewer than two mobility limitations at wave t and two or more at the next observed wave.
<code>ep005_derived_states</code>	easySHARE / derived	<code>employed</code> , <code>retired</code> , <code>unemployed</code> , <code>disabled</code> , and <code>homemaker</code> are indicators built from current job situation.
<code>lfp, hours</code>	derived from <code>ep005_</code> , <code>ep013_mod</code>	<code>lfp</code> = 1 for respondents who are employed or unemployed. <code>hours</code> is weekly hours worked, observed only for employed respondents.
<code>retired_by_65</code>	derived	Person-level retirement outcome used in the age-50–55 baseline exercise. It equals 1 if the respondent is observed retired in any wave up to and including age 65.
<code>post_fornero</code>	derived	Indicator for interviews from 2012 onward, used to separate observations before and after the 2011 Fornero pension reform became binding.
<code>year_turn65, stat_ret_age_at65</code>	derived from cohort and Italian law	<code>year_turn65</code> is the calendar year in which the respondent turns 65. <code>stat_ret_age_at65</code> is the gender- and year-specific statutory old-age retirement age applying in that year.
<code>passes_stat_age_at65</code>	derived from Italian law	Statutory-age-at-65 dummy used in the retire-by-65 models. It equals 1 if a respondent aged exactly 65 would have reached or passed the statutory old-age retirement age under the actual law in the year they turn 65. This captures the Fornero transition but is not a full pension-entitlement measure.

Table A.1: Variable codebook: source and definition

Waves observed	Persons	Share of persons	Cumulative share	Person-waves
9	419	4.9%	4.9%	3,771
8	456	5.4%	10.3%	3,648
7	424	5.0%	15.3%	2,968
6	606	7.1%	22.4%	3,636
5	989	11.6%	34.1%	4,945
4	1,173	13.8%	47.9%	4,692
3	1,135	13.4%	61.2%	3,405
2	1,162	13.7%	74.9%	2,324
1	2,135	25.1%	100.0%	2,135
Total	8,499	100.0%		31,524

Table A.2: Panel participation: persons by number of easySHARE waves observed (Italy, waves 1-9)

Stratum	Total	Male		Female	
	N	N	%	N	%
Education (ISCED)					
Low (ISCED 0-2)	5,831	2,599	44.6%	3,232	55.4%
Medium (ISCED 3-4)	1,843	894	48.5%	949	51.5%
High (ISCED 5-6)	680	353	51.9%	327	48.1%
Migration background					
Italian-born	8,315	3,842	46.2%	4,473	53.8%
Foreign-born	120	35	29.2%	85	70.8%
Urbanicity					
Urban	1,631	749	45.9%	882	54.1%
Non-urban	6,323	2,912	46.1%	3,411	53.9%

Table A.3: Gender Balance across Education (ISCED), Migration Background, and Urbanicity

Variable	Male			Female		
	Mean	SD	N	Mean	SD	N
Main variables						
Age (years)	68.61	9.48	14,250	67.57	9.93	17,274
Max grip strength (kg)	39.81	9.90	12,497	24.99	6.76	14,478
BMI	26.72	3.57	12,903	25.96	4.54	15,570
Self-rated health (1=excellent, 5=poor)	3.21	1.04	14,224	3.36	1.03	17,243
Chronic conditions (count)	1.18	1.19	13,074	1.11	1.18	15,885
Mobility limitations (0-10)	0.53	0.96	13,047	0.76	1.09	15,863
ADL limitations (0-6)	0.21	0.80	13,057	0.29	0.91	15,871
IADL limitations (0-7)	0.11	0.50	13,057	0.13	0.54	15,871
EURO-D depression score (0-12)	2.29	2.32	11,309	3.29	2.68	13,873
Immediate recall (0-10)	4.64	1.81	12,551	4.66	1.83	15,329
Delayed recall (0-10)	3.10	1.94	12,585	3.18	2.01	15,363
Rockwood frailty index (0-1)	0.20	0.15	11,463	0.25	0.17	14,077
In labour force (0/1)	0.24	0.43	14,250	0.16	0.36	17,274
Retired (0/1)	0.70	0.46	13,007	0.41	0.49	15,789
Weekly hours worked (if employed)	40.39	11.92	2,530	34.16	11.46	2,126
Education (ISCED, share)						
Low (ISCED 0-2)	0.68	—	8,893	0.72	—	11,376
Medium (ISCED 3-4)	0.23	—	2,995	0.21	—	3,320
High (ISCED 5-6)	0.09	—	1,110	0.07	—	1,097
Migration background (share)						
Italian-born	0.99	—	14,067	0.98	—	16,879
Foreign-born	0.01	—	122	0.02	—	314
Urbanicity (share)						
Urban	0.18	—	2,245	0.18	—	2,744
Non-urban	0.82	—	10,107	0.82	—	12,278

Table A.4: Pooled Descriptive Statistics by Gender

Wave	Year	N	Mean age	Gender		Education (ISCED)			Health					Labour	
				Male	Female	Low	Medium	High	SRH (1-5)	Chronic	Grip (kg)	EURO-D	Frailty	Employed	Retired
1	2004	2,489	64.9	45%	55%	78%	17%	5%	3.26	1.09	31.3	2.85	0.23	18%	56%
2	2006	148	66.9	44%	56%	65%	23%	12%	3.32	0.91	32.2	2.61	0.20	16%	63%
2	2007	2,763	65.8	46%	54%	74%	20%	6%	3.31	1.16	33.1	2.78	0.23	18%	55%
3	2008	319	67.4	45%	55%	—	—	—	3.37	—	31.7	—	—	—	—
3	2009	2,174	67.1	46%	54%	—	—	—	3.40	—	32.6	—	—	—	—
4	2011	3,502	67.0	46%	54%	71%	23%	6%	3.25	1.14	32.7	2.73	0.23	19%	54%
5	2013	4,645	67.3	46%	54%	71%	20%	8%	3.29	1.13	32.1	3.00	0.24	21%	53%
6	2015	5,199	67.4	46%	54%	69%	23%	8%	3.25	1.08	32.7	2.78	0.21	22%	51%
7	2017	4,508	69.1	45%	55%	69%	23%	8%	3.25	1.14	31.6	3.02	0.25	20%	55%
8	2019	795	71.6	44%	56%	69%	24%	7%	3.35	1.29	29.4	2.90	0.23	16%	56%
8	2020	1,365	70.9	44%	56%	69%	21%	9%	3.32	1.16	30.1	2.90	0.23	18%	55%
9	2021	624	72.1	43%	57%	65%	27%	8%	3.37	1.37	28.7	3.09	0.24	14%	58%
9	2022	2,993	72.2	43%	57%	67%	24%	9%	3.34	1.22	30.2	2.68	0.21	13%	59%

Table A.5: Descriptive Statistics by Wave

The sample is an unbalanced panel. Table A.2 makes explicit that only a small minority of Italian respondents are observed in all nine waves, while about half are observed in four waves or fewer. This matters for the person-fixed-effect estimates in Section 6: respondents with only one observation contribute to pooled prediction exercises, but not to within-person identification.

A.2 Further figures for describing Italian ageing and health

Figure A.1 shows whether the age-health gradient differs by education; it is the education analogue of the main age-profile plots.

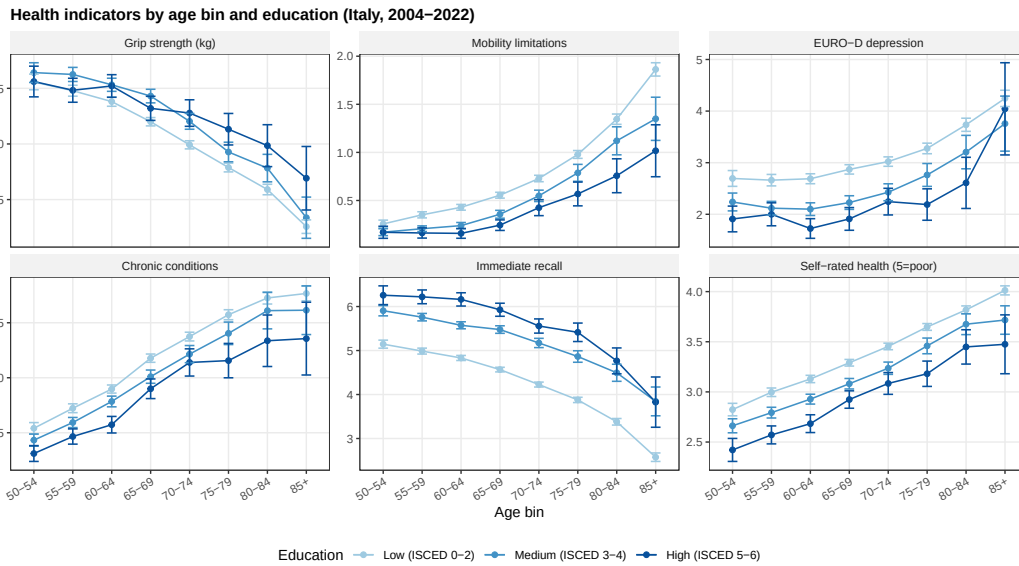


Figure A.1: Age profiles of headline health indicators by education level. Italy, easySHARE waves 1–9

Figure A.2 compares age profiles across interview-year groups to show whether later waves look healthier at the same ages.

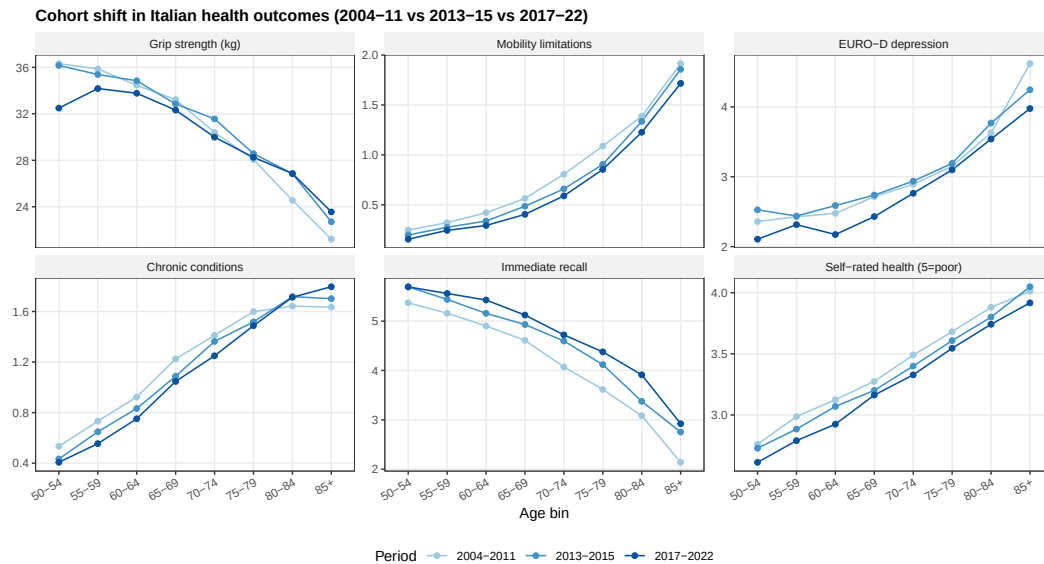


Figure A.2: Age profiles of headline health indicators across interview-year groups. Italy, easySHARE waves 1–9

Appendix B Predicting Two-Year Mobility Onset: Further Results

Table B.1 reports the nested pooled logits and the discrete-time cloglog hazard model used in Section 3; it shows how prediction changes as socio-economic and lagged-health variables are added. Table B.2 reports held-out test performance for the logit and machine-learning benchmarks using respondent-level train/test splits.

	Logit: Demographics	Logit: + SES	Logit: + Lagged health	Cloglog hazard
Age	0.049 (0.041)	0.071 (0.044)	0.067 (0.052)	
Age ²	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	
Female	0.517*** (0.055)	0.509*** (0.060)	0.029 (0.094)	−0.023 (0.088)
ISCED medium		−0.276*** (0.082)	−0.091 (0.091)	−0.091 (0.088)
ISCED high		−0.893*** (0.165)	−0.607*** (0.177)	−0.580*** (0.176)
Income Q2		−0.172** (0.077)	−0.052 (0.089)	−0.051 (0.083)
Income Q3		−0.411*** (0.089)	−0.178* (0.101)	−0.145 (0.095)
Income Q4 (top)		−0.457*** (0.089)	−0.111 (0.102)	−0.113 (0.097)
Foreign-born		0.002 (0.239)	−0.080 (0.273)	−0.048 (0.260)
Prior chronic conditions			0.183*** (0.030)	0.163*** (0.028)
Prior EURO-D			0.112*** (0.014)	0.097*** (0.013)
Prior grip strength			−0.034*** (0.005)	−0.031*** (0.005)
Prior BMI			0.069*** (0.008)	0.057*** (0.007)
Prior immediate recall			−0.052** (0.022)	−0.049** (0.021)
Num.Obs.	15 352	13 312	12 017	11 905
Log.Lik.	−4809.111	−4126.284	−3370.402	−3193.890
McFadden pseudo-R ²	0.095	0.110	0.132	0.119

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.1: Predicting 2-year mobility onset: nested pooled logits and discrete-time cloglog hazard (Italy)

Model	AUC	Brier	Calibration_slope
Logit (m3)	0.763	0.078	0.979
Random forest	0.752	0.078	0.943
XGBoost	0.756	0.079	0.924

Table B.2: Held-out test metrics: logit, random forest, and XGBoost on 2-year mobility onset

Appendix C A Composite Health Index: Further Results

Table C.1 reports the PCA loadings, which show how each health indicator contributes to the first principal component. Table C.2 validates the composite index by comparing mobility-onset prediction using the full health battery, Health PC1, and Rockwood frailty.

Indicator	PC1	PC2	PC3
grossmotor	-0.410	0.119	-0.103
mobilityind	-0.393	0.018	-0.266
adla	-0.367	0.313	0.268
lgmuscle	-0.353	-0.018	-0.278
finemotor	-0.338	0.308	0.271
iadla	-0.276	0.238	0.366
eurod	-0.243	-0.174	-0.173
chronic_mod	-0.215	-0.149	-0.375
recall_1	0.208	0.554	-0.320
maxgrip	0.203	0.182	0.020
recall_2	0.175	0.583	-0.310
bmi	-0.091	-0.058	-0.439

Table C.1: PCA loadings for the first three components (12 Italian easySHARE health indicators)

Model	AUC
Demographics + 9 health variables	0.773
Demographics + PCA PC1	0.775
Demographics + Rockwood frailty	0.777

Table C.2: Validation of composite indices: AUC for 2-year mobility onset (same covariate base)

Appendix D Documenting Health Inequality: Further Results

Table D.1 reports bootstrap confidence intervals for health gaps across inequality dimensions and age bins; it supplies the numerical counterpart to the inequality figures in Section 5.

Age bin	Male - Female	ISCED high - low	Italian-born - Foreign-born	Urban - Non-urban
80-84	1.15 [0.90, 1.40]	1.43 [1.02, 1.77]	-0.68 [-1.26, 0.11]	-0.08 [-0.48, 0.27]
60-64	0.62 [0.53, 0.72]	0.77 [0.56, 0.94]	0.01 [-0.36, 0.46]	-0.15 [-0.31, 0.01]
75-79	1.01 [0.83, 1.17]	1.21 [0.92, 1.50]	-0.12 [-0.88, 0.76]	-0.25 [-0.53, 0.01]
85+	1.52 [1.08, 1.96]	1.06 [-0.10, 2.22]	0.71 [-1.21, 1.98]	-0.22 [-0.93, 0.45]
50-54	0.47 [0.33, 0.60]	0.56 [0.33, 0.76]	0.22 [-0.07, 0.55]	-0.13 [-0.36, 0.06]
65-69	0.64 [0.54, 0.75]	0.94 [0.80, 1.07]	-0.09 [-0.44, 0.35]	-0.07 [-0.23, 0.09]
55-59	0.56 [0.46, 0.65]	0.64 [0.46, 0.80]	0.15 [-0.31, 0.68]	-0.10 [-0.26, 0.04]
70-74	0.81 [0.70, 0.92]	0.86 [0.65, 1.05]	-0.17 [-0.57, 0.29]	-0.02 [-0.21, 0.17]

Table D.1: Health gaps across inequality dimensions (bootstrap 95% CIs, R=500)

Appendix E Health and Economic Outcomes: Further Results and Analysis

E.1 Health and labour force decisions: numerical results

Tables E.1 and E.2 report held-out predictive performance for labour-force participation and weekly hours. Table E.3 reports the pooled logit and OLS regression coefficients behind these prediction exercises.

Horizon	Outcome	AUC			Brier			N
		Logit	RF	XGBoost	Logit	RF	XGBoost	
h=0	LFP	0.866	0.871	0.879	0.141	0.137	0.133	4011
h=1	LFP	0.854	0.851	0.866	0.131	0.131	0.125	3076
h=2	LFP	0.887	0.873	0.883	0.103	0.109	0.104	2449
h=3	LFP	0.915	0.902	0.914	0.076	0.082	0.077	1902

Table E.1: Held-out predictive performance: pooled logit, random forest and XGBoost for LFP at horizons 0–3 waves (Italy, 50–70). Same predictors as the person-FE regressions in Table E.4: health PC1, Rockwood frailty, age, female, ISCED, income quartile, migration.

Horizon	Outcome	RMSE			R^2			N
		OLS	RF	XGBoost	OLS	RF	XGBoost	
h=0	Hours — LFP	10.75	11.19	10.95	0.086	0.010	0.053	1237
h=1	Hours — LFP	9.57	9.91	9.86	0.068	0.001	0.011	467
h=2	Hours — LFP	9.23	9.74	9.51	0.100	-0.000	0.046	321
h=3	Hours — LFP	9.56	9.99	10.18	0.081	-0.003	-0.042	242

Table E.2: Held-out predictive performance: pooled OLS, random forest and XGBoost for weekly hours conditional on employment, at horizons 0–3 waves.

Variable	LFP logit				Hours OLS			
	h=0	h=1	h=2	h=3	h=0	h=1	h=2	h=3
Intercept	16.841*** (0.415)	15.039*** (0.470)	17.640*** (0.621)	21.961*** (0.920)	40.784*** (2.926)	52.387*** (4.895)	43.999*** (6.799)	59.048*** (8.326)
Health PC1	0.132*** (0.033)	0.091** (0.038)	0.067 (0.044)	0.079 (0.057)	0.856*** (0.264)	0.263 (0.352)	0.780 (0.500)	0.803* (0.424)
Rockwood frailty	0.172 (0.363)	-0.351 (0.432)	-0.461 (0.516)	0.201 (0.670)	3.898 (2.783)	0.748 (4.104)	1.508 (5.193)	-4.100 (5.709)
Age	-0.285*** (0.007)	-0.269*** (0.008)	-0.320*** (0.010)	-0.405*** (0.016)	-0.047 (0.048)	-0.211*** (0.081)	-0.118 (0.118)	-0.382** (0.149)
Female	-1.359*** (0.061)	-1.153*** (0.070)	-1.129*** (0.087)	-1.216*** (0.112)	-6.107*** (0.422)	-6.670*** (0.647)	-6.543*** (0.816)	-4.119*** (0.945)
ISCED: Medium (ISCED 3-4)	0.753*** (0.063)	0.510*** (0.074)	0.696*** (0.092)	0.607*** (0.118)	-0.408 (0.456)	0.066 (0.710)	0.929 (0.877)	0.540 (1.036)
ISCED: High (ISCED 5-6)	1.528*** (0.097)	1.279*** (0.112)	1.556*** (0.135)	1.619*** (0.171)	-1.270** (0.593)	-0.833 (0.897)	0.369 (1.130)	-0.201 (1.301)
Income quartile: Q2 (dec 3-5)	-0.331*** (0.088)	-0.075 (0.107)	-0.083 (0.131)	-0.244 (0.170)	1.252* (0.739)	0.787 (1.148)	1.637 (1.311)	-0.124 (1.578)
Income quartile: Q3 (dec 6-7)	-0.274*** (0.092)	-0.049 (0.110)	-0.105 (0.136)	-0.280 (0.177)	0.753 (0.727)	-0.502 (1.118)	0.876 (1.347)	1.673 (1.572)
Income quartile: Q4 (dec 8-10)	0.256*** (0.080)	0.335*** (0.096)	-0.016 (0.119)	0.105 (0.153)	2.394*** (0.626)	0.276 (0.973)	1.907* (1.130)	0.622 (1.380)
Migration: Foreign-born	0.480** (0.208)	0.287 (0.233)	0.826*** (0.284)	0.740* (0.430)	-1.696 (1.505)	-3.015 (2.100)	1.276 (3.187)	-6.493*** (2.407)

Note:

LFP columns report logit coefficients; weekly-hours columns report OLS coefficients conditional on employment. Models are estimated on the training split used for the held-out prediction exercise. Standard errors are in parentheses. Stars: * p<0.10, ** p<0.05, *** p<0.01.

Table E.3: Pooled econometric regressions for labour-force participation and weekly hours, horizons 0–3

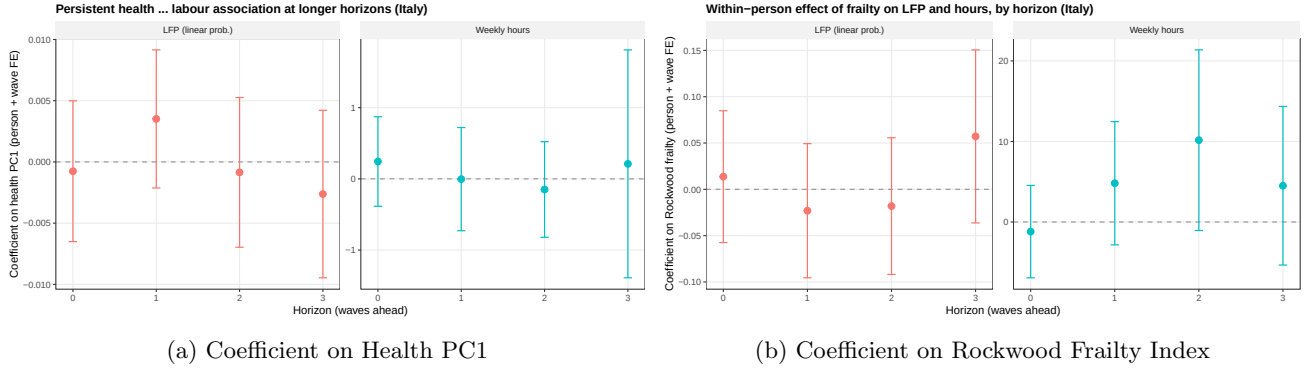
E.2 Health and labour force decisions: within-person FE model

The between-person estimator does not separate the role of health *itself* from everything correlated with health at the individual level. To check whether the gradient survives once permanent person-level heterogeneity is absorbed, we estimate

$$Y_{i,t+h} = \alpha_i + \delta_t + \beta_h \cdot \text{PC1}_{i,t} + \gamma_1 \text{age}_{i,t} + \gamma_2 \text{age}_{i,t}^2 + \varepsilon_{i,t},$$

with person and wave fixed effects, person-clustered SE, ages 50–70. Table E.4 shows that the within-person LFP coefficient on PC1 is economically tiny and statistically indistinguishable from zero at every horizon; the same holds for weekly hours. Replacing PC1 with the 30-item Rockwood frailty index gives the same broad pattern,

apart from one isolated weakly significant hours coefficient at $h = 2$, consistent with the near-equivalence of the two composites documented in Section 4. The point estimates are visualised in Figure E.1.



Note: Italy 50–70. Person and wave fixed effects, person-clustered SE.

Figure E.1: Within-person effect of health on labour force variables

Variable	LFP				Weekly hours			
	h = 0	h = 1	h = 2	h = 3	h = 0	h = 1	h = 2	h = 3
Health PC1 specification								
Health PC1	-0.001 (0.003)	0.004 (0.003)	-0.001 (0.003)	-0.003 (0.003)	0.244 (0.321)	-0.004 (0.370)	-0.150 (0.343)	0.211 (0.817)
Age	0.009 (0.030)	0.033 (0.032)	-0.030 (0.034)	-0.238*** (0.041)	3.905* (2.061)	7.070* (3.629)	-7.533* (4.414)	6.011 (3.816)
Age ²	-0.001*** (0.000)	-0.000* (0.000)	0.001*** (0.000)	0.002*** (0.000)	-0.020* (0.012)	-0.052** (0.021)	0.049* (0.029)	-0.016 (0.030)
Num.Obs.	13,461	10,480	8,265	6,291	4,193	1,576	1,062	798
Rockwood frailty specification								
Rockwood frailty	0.014 (0.036)	-0.023 (0.037)	-0.018 (0.038)	0.057 (0.048)	-1.195 (2.925)	4.816 (3.906)	10.160* (5.721)	4.502 (5.023)
Age	0.030 (0.028)	0.037 (0.030)	-0.051* (0.031)	-0.226*** (0.039)	4.172** (1.937)	9.657*** (3.463)	-7.793** (3.769)	5.551 (3.809)
Age ²	-0.001*** (0.000)	-0.000* (0.000)	0.001*** (0.000)	0.002*** (0.000)	-0.024** (0.011)	-0.065*** (0.020)	0.042* (0.024)	-0.018 (0.029)
Num.Obs.	14,999	11,517	9,046	6,839	4,563	1,722	1,144	851

Notes: Each column is a separate person- and wave-fixed-effects regression for the listed outcome and horizon. The first block uses Health PC1 as the health measure; the second uses Rockwood frailty. Coefficient rows report standard errors clustered at the person level in parentheses below estimates; Num.Obs. reports the estimation sample size. LFP models are linear-probability models; weekly-hours models are OLS conditional on observed hours. h is the horizon in SHARE waves ahead. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E.4: Person-FE regressions of labour outcomes on within-person health changes.

E.3 Definition of the health shock in the LP-DiD and robustness

The LP-DiD exercise requires a dated deterioration event. A continuous health index is less suitable for this purpose because it moves at every wave and can mix genuine deterioration with small reporting or measurement changes. We therefore define the event as the first observed wave in which a respondent experiences a clear worsening in at least one major health domain. In the baseline specification,

$$\text{health_shock}_{it} = 1$$

if, relative to the previous observed wave, any of the following holds:

$$\Delta \text{chronic conditions}_{it} \geq 1, \quad \Delta \text{mobility limitations}_{it} \geq 2, \quad \text{EURO-D}_{i,t-1} < 4 \text{ and } \text{EURO-D}_{it} \geq 4.$$

The first component captures new morbidity; the second captures a sizeable functional deterioration; the third captures crossing the standard EURO-D depression-risk threshold. The definition is deliberately transparent and survey-based. It is not meant to identify a clinical diagnosis, but a dated worsening in morbidity, functioning, or mental health that could plausibly affect work.

This choice follows the broader health-shock labour-supply literature in spirit rather than copying a single canonical formula. Existing studies commonly date shocks using new diagnoses or acute events such as heart attack, stroke, or cancer (Coile, 2004; Trevisan & Zantomio, 2016); others use sharp deterioration in self-assessed health or new health conditions (Lenhart, 2019). Because the easySHARE panel used here contains a chronic-condition count rather than separate incident diagnoses for infarction, stroke, and cancer, the baseline definition combines morbidity, functional limitation, and depression indicators.

As a robustness check, Table E.5 re-estimates the pooled LP-DiD using the self-rated-health approach. The alternative shock equals one when `sphus` worsens by two or more categories since the previous observed wave, corresponding to a severe deterioration in self-assessed health. This is closer to Lenhart (2019), who uses declines in self-reported health status as the main health-shock measure and distinguishes more severe declines from milder ones. The table checks whether the weak average labour-supply response depends on the baseline shock definition.

Outcome	Shock definition	k = -3	N	k = -2	N	k = 0	N	k = 1	N	k = 2	N	k = 3	N
LFP	Baseline composite	-0.015 (0.019)	3,642	-0.007 (0.014)	5,578	-0.020* (0.011)	7,832	0.000 (0.014)	5,578	-0.011 (0.016)	3,642	0.031 (0.022)	2,131
LFP	SRH worsens by ≥ 2	-0.039 (0.029)	1,346	-0.002 (0.021)	2,049	0.001 (0.019)	2,859	0.002 (0.022)	2,049	0.018 (0.028)	1,346	0.022 (0.039)	799
Weekly hours	Baseline composite	-0.224 (1.203)	534	1.365 (0.936)	790	0.474 (0.905)	994	-0.462 (1.093)	655	0.382 (1.422)	505	-0.578 (2.535)	248
Weekly hours	SRH worsens by ≥ 2	1.291 (1.977)	208	-1.189 (1.836)	307	-1.100 (1.616)	370	2.889* (1.612)	247	1.513 (1.761)	183	-0.152 (2.969)	99

Notes: Baseline composite shock is the main definition: at least one additional chronic condition, at least two additional mobility limitations, or EURO-D crossing from below 4 to 4 or above. The alternative self-rated-health (SRH) shock equals one when self-rated health worsens by two or more categories since the previous observed wave. Outcomes are Δ LFP and Δ weekly hours relative to the previous wave. All regressions control for age, age squared, and wave fixed effects, with standard errors clustered at the person level. $k = -3$ and $k = -2$ are pre-shock placebo horizons; $k = 0, 1, 2, 3$ are contemporaneous and post-shock horizons; $k = -1$ is the omitted reference. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E.5: Robustness of pooled LP-DiD estimates to an alternative health-shock definition.

E.4 Heterogeneity in the labour-supply response to health shocks

Table E.6 reports the LP-DiD estimates around the first observed health shock for both labour-force participation and weekly hours conditional on employment. The row structure mirrors the main heterogeneity figure: one pooled row followed by age-group $\times$ ISCED cells.

Age group	ISCED	k = -3	N	k = -2	N	k = 0	N	k = 1	N	k = 2	N	k = 3	N
Labour-force participation													
Pooled	All	-0.015 (0.019)	3,642	-0.007 (0.014)	5,578	-0.020* (0.011)	7,832	0.000 (0.014)	5,578	-0.011 (0.016)	3,642	0.031 (0.022)	2,131
50-59	Low	-0.062 (0.079)	210	-0.114*** (0.036)	516	-0.042* (0.025)	1,051	-0.014 (0.028)	965	-0.024 (0.030)	759	0.020 (0.037)	506
50-59	Medium	-0.188*** (0.050)	137	-0.252*** (0.048)	312	-0.055* (0.031)	572	0.014 (0.033)	524	0.063* (0.035)	408	0.060 (0.044)	277
50-59	High	-0.232** (0.098)	24	-0.247*** (0.075)	70	-0.038 (0.038)	146	0.065 (0.044)	128	0.148** (0.060)	90	0.066 (0.088)	57
60-64	Low	-0.043 (0.037)	792	-0.051* (0.028)	1,065	-0.024 (0.020)	1,461	0.027 (0.021)	1,176	-0.008 (0.025)	839	0.076** (0.038)	478
60-64	Medium	-0.020 (0.058)	400	-0.001 (0.042)	521	-0.030 (0.032)	659	-0.051 (0.035)	526	-0.133*** (0.038)	368	-0.116** (0.046)	202
60-64	High	0.017 (0.163)	92	-0.150 (0.094)	142	-0.093 (0.063)	196	-0.141** (0.058)	150	-0.208*** (0.062)	97	-0.220*** (0.071)	41
65-70	Low	0.030 (0.026)	1,293	0.075*** (0.019)	1,584	0.029** (0.014)	1,980	0.048** (0.020)	924	-0.007 (0.037)	267	-0.215*** (0.049)	4
65-70	Medium	-0.005 (0.045)	524	0.103*** (0.036)	620	-0.026 (0.027)	706	-0.069* (0.036)	322	-0.122** (0.058)	88	-0.210*** (0.047)	1
65-70	High	0.098 (0.061)	148	0.079 (0.061)	185	-0.044 (0.051)	208	-0.100 (0.066)	91	-0.058 (0.104)	23	—	—
Weekly hours conditional on employment													
Pooled	All	-0.224 (1.203)	534	1.365 (0.936)	790	0.474 (0.905)	994	-0.462 (1.093)	655	0.382 (1.422)	505	-0.578 (2.535)	248
50-59	Low	-0.013 (4.826)	46	-1.033 (1.868)	116	-0.443 (2.161)	227	-1.513 (2.200)	125	0.522 (2.286)	140	0.008 (4.837)	62
50-59	Medium	0.182 (2.109)	62	2.837 (1.792)	86	1.886 (1.675)	200	1.186 (1.394)	149	-0.798 (1.764)	138	-0.050 (4.067)	66
50-59	High	-4.919 (3.847)	11	2.628* (1.506)	27	-2.042 (1.866)	75	1.007 (2.115)	46	-0.087 (2.256)	49	0.396 (4.596)	24
60-64	Low	-2.788 (3.250)	104	2.016 (2.013)	105	4.819** (2.310)	139	-1.009 (1.943)	60	1.290 (2.896)	20	12.848 (9.208)	3
60-64	Medium	1.655 (1.091)	125	0.528 (0.997)	132	-0.988 (1.149)	166	-1.497 (1.653)	66	0.077 (1.940)	30	-7.823** (3.867)	6
60-64	High	-1.033 (6.158)	43	-3.457 (5.705)	54	5.166* (2.793)	78	-4.460 (4.078)	34	-0.311 (3.034)	12	7.328 (6.491)	2
65-70	Low	6.718** (3.120)	54	7.070 (4.817)	59	-3.821 (4.020)	40	-1.643 (1.350)	5	—	—	—	—
65-70	Medium	-0.696 (2.748)	64	1.024 (2.416)	61	-6.042** (2.987)	41	0.070 (1.823)	9	12.991*** (4.274)	2	—	—
65-70	High	2.999 (3.704)	24	1.230 (2.259)	29	2.487* (1.483)	21	—	—	—	—	—	—

Notes: Outcomes are Δ LFP and Δ weekly hours relative to the previous wave. Weekly hours are estimated conditional on observed hours worked at the reference and horizon observations. All regressions control for age, age squared, and wave fixed effects. Standard errors, clustered at the person level, are reported in parentheses below each estimate. The pooled row reports the coefficient on the LP-DiD treatment indicator at each event-time horizon. The remaining rows report interaction estimates by age group and ISCED level. $k = -3$ and $k = -2$ are pre-shock placebo horizons; $k = 0, 1, 2, 3$ are contemporaneous and post-shock horizons; $k = -1$ is the omitted reference. N is the horizon-specific estimation sample size. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. Dashes indicate cells omitted by the model because the interaction is not separately identified.

Table E.6: LP-DiD estimates for labour-force participation and weekly hours around the first health shock.

Figure E.2 is the weekly-hours analogue of the main LFP heterogeneity heatmap, using the same age-group $\times$ ISCED cells.

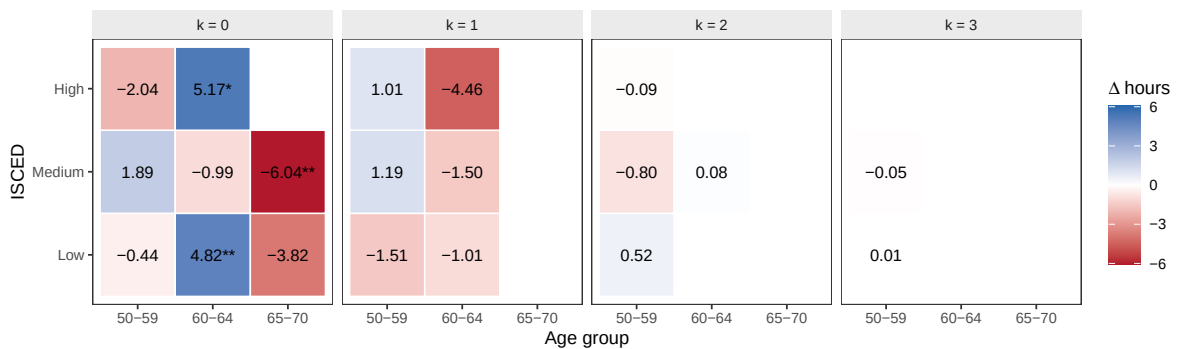


Figure E.2: LP-DiD heterogeneity heatmap for weekly hours by age group and ISCED. Each tile is one (age group $\times$ ISCED) cell at $k = 0, 1, 2, 3$ waves around the first observed health shock; cell colour is the estimated Δ weekly hours conditional on employment and the overlaid number is the point estimate (stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$). Cells with $N < 30$ are omitted from the heatmap. Sample: Italy, ages 50–70. See Table E.6 for full numerical results.

Age group	Income tercile	k = -3	N	k = -2	N	k = 0	N	k = 1	N	k = 2	N	k = 3	N
Labour-force participation													
Pooled	All	-0.015 (0.019)	3,642	-0.007 (0.014)	5,578	-0.020* (0.011)	7,832	0.000 (0.014)	5,578	-0.011 (0.016)	3,642	0.031 (0.022)	2,131
50-59	Bottom	0.036 (0.106)	79	-0.073 (0.060)	198	-0.073** (0.036)	429	-0.014 (0.039)	389	0.014 (0.041)	320	0.074* (0.044)	260
50-59	Middle	-0.127 (0.119)	76	-0.285*** (0.080)	160	-0.062 (0.040)	340	-0.054 (0.038)	318	0.007 (0.040)	276	-0.010 (0.044)	225
50-59	Top	-0.197*** (0.055)	158	-0.258*** (0.044)	331	-0.058** (0.026)	641	-0.010 (0.031)	588	0.047 (0.037)	474	0.043 (0.042)	368
60-64	Bottom	-0.113* (0.058)	362	-0.104** (0.051)	456	-0.034 (0.031)	616	-0.021 (0.031)	473	-0.069* (0.038)	357	-0.015 (0.045)	236
60-64	Middle	-0.007 (0.058)	310	-0.043 (0.042)	398	-0.020 (0.037)	525	0.027 (0.031)	417	0.012 (0.033)	317	0.076* (0.046)	200
60-64	Top	-0.004 (0.052)	505	-0.024 (0.036)	641	-0.041* (0.025)	866	-0.046* (0.027)	685	-0.093*** (0.032)	493	-0.042 (0.041)	297
65-70	Bottom	0.011 (0.039)	566	0.074** (0.032)	660	0.014 (0.022)	789	-0.044 (0.030)	373	-0.097* (0.052)	105	-0.201*** (0.047)	2
65-70	Middle	0.038 (0.036)	559	0.107*** (0.030)	646	0.035 (0.022)	784	0.034 (0.026)	369	-0.038 (0.045)	121	-0.206*** (0.049)	3
65-70	Top	0.020 (0.034)	738	0.099*** (0.026)	853	0.003 (0.020)	1,033	0.021 (0.026)	508	-0.032 (0.047)	126	—	—
Weekly hours conditional on employment													
Pooled	All	-0.224 (1.203)	534	1.365 (0.936)	790	0.474 (0.905)	994	-0.462 (1.093)	655	0.382 (1.422)	505	-0.578 (2.535)	248
50-59	Bottom	7.900 (5.230)	10	-1.038 (5.764)	19	-0.418 (3.450)	89	1.925 (5.653)	24	4.362 (4.241)	36	6.289 (5.159)	35
50-59	Middle	-5.937 (5.330)	18	5.381 (6.892)	17	-0.737 (2.343)	106	-3.181 (2.681)	35	-1.822 (3.220)	51	-3.439 (4.358)	35
50-59	Top	-2.298 (2.363)	59	1.892 (1.804)	69	0.642 (1.481)	309	-0.202 (2.542)	108	-1.176 (2.249)	150	-2.677 (4.091)	83
60-64	Bottom	-3.701 (5.398)	58	1.776 (3.785)	36	2.960 (3.129)	82	1.923 (3.770)	23	2.965 (3.801)	9	-8.695* (4.537)	2
60-64	Middle	-1.223 (4.686)	47	-2.685 (8.330)	30	1.277 (2.053)	72	-2.945 (3.083)	24	5.639* (3.171)	9	17.241 (11.615)	2
60-64	Top	0.524 (1.249)	123	2.313 (1.796)	116	1.511 (1.680)	234	-4.331** (1.699)	62	-4.285* (2.284)	26	-0.693 (5.565)	7
65-70	Bottom	6.537 (5.006)	35	2.138 (3.455)	29	1.137 (3.665)	20	—	—	—	—	—	—
65-70	Middle	2.884 (3.461)	32	14.430 (10.234)	31	-3.601* (1.916)	24	1.913 (1.917)	4	—	—	—	—
65-70	Top	-2.105 (2.927)	65	1.180 (1.850)	62	-5.686* (3.435)	58	1.977 (2.513)	7	11.494*** (4.421)	1	—	—

Notes: Outcomes are Δ LFP and Δ weekly hours relative to the previous wave. Weekly hours are estimated conditional on observed hours worked at the reference and horizon observations. All regressions control for age, age squared, and wave fixed effects. Standard errors, clustered at the person level, are reported in parentheses below each estimate. The pooled row reports the coefficient on the LP-DiD treatment indicator at each event-time horizon. The remaining rows report interaction estimates by age group and income tercile. $k = -3$ and $k = -2$ are pre-shock placebo horizons; $k = 0, 1, 2, 3$ are contemporaneous and post-shock horizons; $k = -1$ is the omitted reference. N is the horizon-specific estimation sample size. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Dashes indicate cells omitted by the model because the interaction is not separately identified.

Table E.7: LP-DiD estimates by age group and income tercile around the first health shock.

Table E.7 keeps the age-group $\times$ income-tercile LP-DiD specification in the appendix as a robustness and comparability check. Figures E.3 and E.4 visualise those income-cell estimates for LFP and weekly hours.

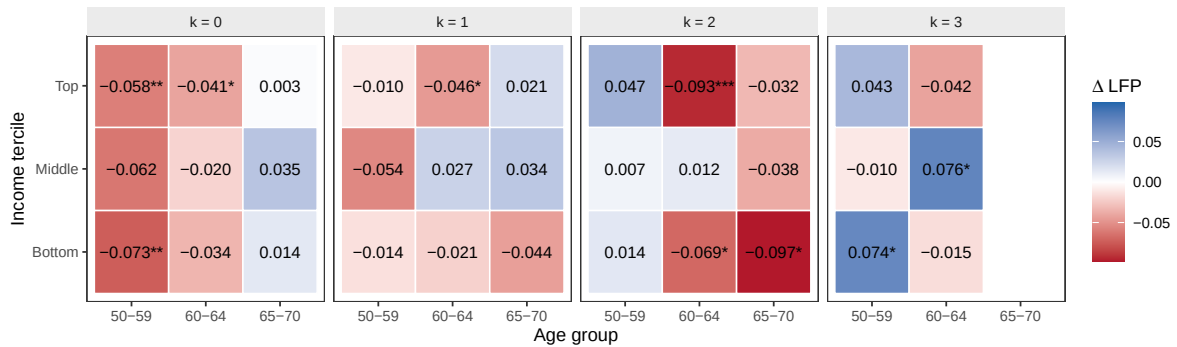


Figure E.3: LP-DiD heterogeneity heatmap for labour-force participation by age group and income tercile. Each tile is one (age group $\times$ income-tercile) cell at $k = 0, 1, 2, 3$ waves around the first observed health shock; cell colour is the estimated Δ LFP and the overlaid number is the point estimate (stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). Cells with $N < 30$ are omitted from the heatmap. Sample: Italy, ages 50–70. See Table E.7 for full numerical results.

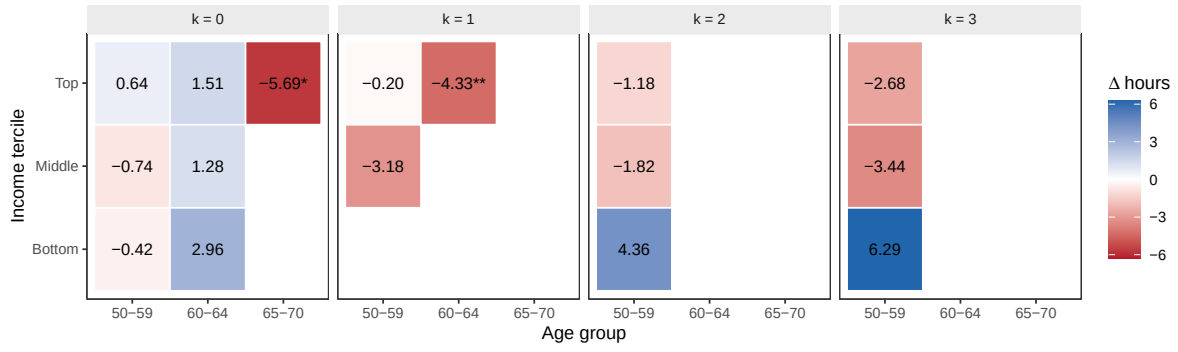


Figure E.4: LP-DiD heterogeneity heatmap for weekly hours by age group and income tercile. Each tile is one (age group $\times$ income-tercile) cell at $k = 0, 1, 2, 3$ waves around the first observed health shock; cell colour is the estimated Δ weekly hours conditional on employment and the overlaid number is the point estimate (stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$). Cells with $N < 30$ are omitted from the heatmap. Sample: Italy, ages 50–70. See Table E.7 for full numerical results.

E.5 Health at 50-55 and later-life retirement

Table E.8 reports the simple counterfactual retirement rates by education group. Table E.9 reports retire-by-65 prediction performance across logit, random forest, and XGBoost. Table E.10 reports the logit coefficients used for the main retirement counterfactuals. Table E.11 combines observed, logit-predicted, and XGBoost-predicted retire-by-65 rates with CF1 and CF2 means by gender and education.

Scenario	Predicted retire-by-65
High-ISCED (observed)	0.119
Low-ISCED (observed)	0.182
Low-ISCED, high-ISCED median-health CF (CF1)	0.190
Low-ISCED, DFL-reweighted to high-ISCED health (CF2)	0.189
Medium-ISCED (observed)	0.186
Medium-ISCED, high-ISCED median-health CF (CF1)	0.190
Medium-ISCED, DFL-reweighted to high-ISCED health (CF2)	0.195

Table E.8: Counterfactual predicted rate of retirement by age 65 (baseline age 50-55)

Model	AUC	Brier	Calibration slope
Logit	0.709	0.132	1.060
Random forest	0.658	0.142	0.581
XGBoost	0.708	0.157	2.959

Table E.9: Held-out predictive performance: logit, random forest, and XGBoost for retire-by-65, age-50-55 baseline (Italy). Predictors include demographics, SES, Health PC1, and a statutory-age-at-65 dummy.

Variable	Logit coefficient
Intercept	-132.642 (115.420)
Age	4.760 (4.389)
Age squared	-0.043 (0.042)
Female	-0.918*** (0.147)
ISCED: Medium (ISCED 3-4)	-0.053 (0.150)
ISCED: High (ISCED 5-6)	-0.816*** (0.250)
Income quartile: Q2 (dec 3-5)	0.084 (0.243)
Income quartile: Q3 (dec 6-7)	0.480* (0.245)
Income quartile: Q4 (dec 8-10)	0.831*** (0.206)
Migration: Foreign-born	-0.363 (0.548)
Passes statutory age at 65	1.435*** (0.298)
Health PC1	0.093 (0.067)

Note:

Outcome is whether the respondent is observed retired by age 65. The statutory-age-at-65 dummy equals one if a respondent aged exactly 65 would have reached or passed the gender/year-specific statutory old-age retirement age under the law in the calendar year when they turn 65. Standard errors are in parentheses. Stars: * p<0.10, ** p<0.05, *** p<0.01.

Table E.10: Logit regression coefficients for retire-by-65, age-50-55 baseline

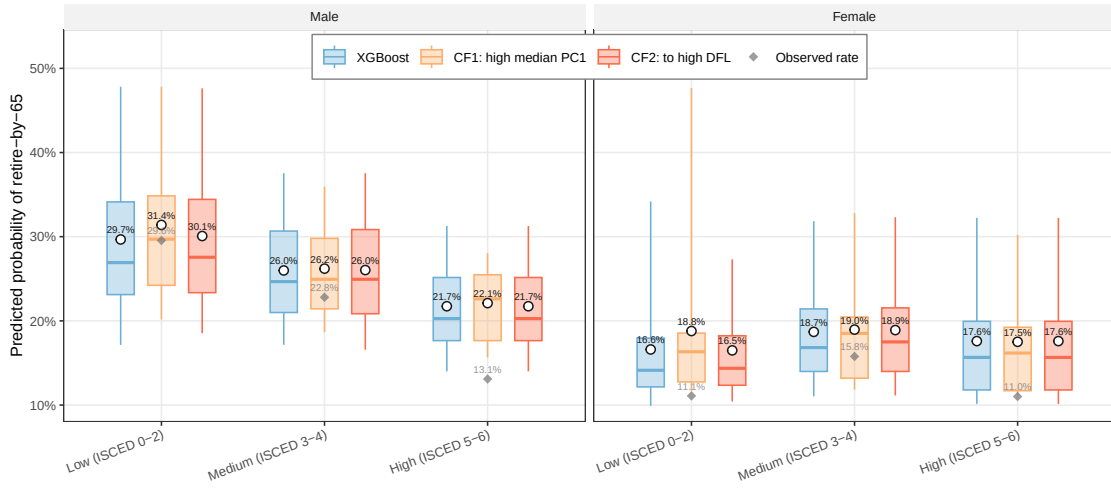
Gender	ISCED	N	Observed	Logit			XGBoost		
				Predicted	CF1	CF2	Predicted	CF1	CF2
Male	Low (ISCED 0-2)	335	0.296	0.261	0.267	0.270	0.297	0.314	0.301
Male	Medium (ISCED 3-4)	228	0.228	0.263	0.264	0.274	0.260	0.262	0.260
Male	High (ISCED 5-6)	84	0.131	0.176	0.177	0.176	0.217	0.221	0.217
Female	Low (ISCED 0-2)	541	0.111	0.133	0.142	0.129	0.166	0.188	0.165
Female	Medium (ISCED 3-4)	336	0.158	0.134	0.139	0.139	0.187	0.190	0.189
Female	High (ISCED 5-6)	118	0.110	0.078	0.079	0.078	0.176	0.175	0.176

Note:

Observed is the cell mean of the retire-by-65 indicator. Predicted is the unweighted cell mean of fitted probabilities from the indicated model. CF1 is the unweighted cell mean of individual counterfactual predicted probabilities after assigning each respondent the high-ISCED median Health PC1. CF2 is the DFL-weighted cell mean of fitted probabilities using source-to-high health-distribution weights; high-ISCED cells are the target/reference and retain weight 1.

Table E.11: Subgroup decomposition: observed, logit-predicted, and XGBoost-predicted CF1 and CF2 retire-by-65 rates by gender and education (baseline age 50-55, Italy).

Figure E.5 repeats the main retirement counterfactual distribution plot using XGBoost predictions rather than the logit. It checks whether the counterfactual patterns are specific to the parametric model.



Note: XGBoost version of Figure 12. CF1: high-ISCED median health; CF2: DFL reweighted. Each box spans the 25th–75th percentiles, the line inside the box is the median, and the whiskers span the 5th–95th percentiles; white circles show means; black diamonds show observed rates. Numerical results in Table E.11.

Figure E.5: Distribution of XGBoost-predicted and counterfactual retire-by-65 probabilities by ISCED and gender.

E.6 Extension: further analysis of the 2011 Fornero reform

We use the Fornero reform as a policy shock to old-age labour supply. The sample is Italy, ages 55–66. Baseline health tercile is fixed at each person’s earliest valid Health PC1 observation, so the health group is predetermined relative to post-2012 labour-force participation. We report two linear-probability specifications. The first is a descriptive pre/post model:

$$\text{LFP}_{i,t} = \alpha_{a(i,t)} + \beta \cdot \text{Post}_t + \sum_{g \in \{\text{middle}, \text{top}\}} \gamma_g \cdot \text{Post}_t \times \text{HealthTercile}_{i,g} + X'_{i,t} \delta + u_{i,t},$$

where $\text{Post}_t = 1[\text{interview year} \geq 2012]$, $\alpha_{a(i,t)}$ are single-year age fixed effects, and $X_{i,t}$ includes gender, education, income quartile, and migration background. The second is an exposure-based fixed-effects model. It defines $\text{Delayed}_{i,t} = 1$ for post-2012 observations where the respondent would have passed the old statutory age threshold, 60 for women and 65 for men, but had not yet passed the actual gender/year-specific statutory age after the reform. This model includes person and interview-year fixed effects:

$$\text{LFP}_{i,t} = \alpha_i + \lambda_t + \theta \cdot \text{Delayed}_{i,t} + \sum_{g \in \{\text{middle}, \text{top}\}} \kappa_g \cdot \text{Delayed}_{i,t} \times \text{HealthTercile}_{i,g} + u_{i,t}.$$

Standard errors are clustered at the person level in both models.

Table E.13 reports both specifications. In the pre/post column, the post-2012 coefficient is the shift for the bottom baseline-health tercile. In the exposure-based column, the delayed-by-Fornero coefficient is the within-person change for the bottom baseline-health tercile when old-law eligibility would have arrived but actual-law eligibility had not. The interaction terms show how much larger or smaller these shifts are for the middle and top health terciles.

Table E.12:

	Pre/post LPM	Exposure FE LPM
Post-Fornero (≥ 2012)	0.152*** (0.019)	
Middle health tercile	-0.009 (0.017)	
Top health tercile	0.047** (0.020)	
Post $\times$ Middle health	0.052** (0.025)	
Post $\times$ Top health	0.026 (0.025)	
Delayed by Fornero		0.053* (0.031)
Delayed $\times$ Middle health		-0.007 (0.036)
Delayed $\times$ Top health		-0.079** (0.037)
Female	-0.222*** (0.013)	
Foreign-born	0.101* (0.057)	
Num.Obs.	8714	8714
Std.Errors	by: mergeid	by: mergeid
FE: mergeid		X
FE: int_year		X

* p < 0.1, ** p < 0.05, *** p < 0.01

Table E.13: Fornero reform (2011): pre/post and exposure-based LPM estimates for LFP, ages 55-66

Appendix F Methodology Notes

F.1 Choice of indicators for the PCA composite health index

The PCA index is intended to summarise general health in one variable that can be used in the labour-market regressions. The inputs are 12 standardised SHARE measures that cover physical strength, cognition, functional limitations, chronic disease, mental health, and body composition:

- ◇ grip strength and large-muscle limitations;
- ◇ immediate and delayed word recall;
- ◇ ADL, IADL, mobility, gross-motor, and fine-motor limitations;
- ◇ chronic-condition count and EURO-D depression score;
- ◇ body-mass index.

We keep these variables because they are direct health states and are measured on continuous or count-like scales. We exclude variables that are mainly behaviours or utilisation, such as smoking, doctor visits, and hospitalisation, and we do not add individual EURO-D symptoms on top of the EURO-D total score. Those additions would duplicate information already in the index and would reduce the complete-case sample. Self-rated health is also

excluded from the PCA because it is an ordinal global assessment rather than a direct component of physical or cognitive capacity.

The first principal component is oriented so that higher values mean better health. It captures about 37% of the pooled variance in the 12 indicators, with positive loadings on grip and cognition and negative loadings on limitations, chronic conditions, and depression. Two checks support using it as the baseline scalar health measure: it is strongly correlated with the independently constructed Rockwood frailty index ($\rho = -0.82$), and replacing the raw health battery with PC1 in the mobility-onset prediction model does not reduce predictive performance (Table C.2).

F.2 Counterfactual retirement exercises: CF1 and CF2

The retirement counterfactuals in Section 6.3 use the age-50–55 baseline sample. For each person, we keep the first observation in that age range and define the outcome as whether the person is observed retired by age 65. The reference model is a logit for retire-by-65:

$$\Pr(R_i = 1) = \Lambda(\alpha + f(\text{age}_i) + \text{gender}_i + \text{ISCED}_i + \text{income quartile}_i + \text{migration}_i + \theta S_i + \beta H_i),$$

where H_i is Health PC1 at baseline, $f(\text{age})$ includes age and age squared, and S_i is the statutory-age-at-65 dummy. Because everyone is observed at ages 50–55 at baseline, S_i is defined at the target age rather than at baseline: it equals one if a respondent aged exactly 65 would have reached or passed the statutory old-age retirement age under the actual gender/year-specific law in the calendar year when they turn 65. The schedule is gender- and year-specific: before 2012 it uses age 65 for men and age 60 for women; after the 2011 Fornero reform it applies the transitional increases and age 67 from 2019 onward. Since easySHARE does not identify pension sector, this is an approximate statutory old-age eligibility measure rather than a full pension-entitlement calculation. The model is not used as a causal estimate of health. It is used as a disciplined prediction rule for asking how predicted retirement rates would change if the health distribution were altered while the estimated relationship between covariates and retirement were held fixed.

CF1 is the *scalar health counterfactual*. Let $\tilde{H}^{\text{high}}$ be the median Health PC1 among high-ISCED respondents at ages 50–55. For each respondent we replace H_i with $\tilde{H}^{\text{high}}$ and re-predict retire-by-65 from the same logit:

$$\hat{p}_i^{CF1} = \widehat{\Pr}(R_i = 1 \mid H_i = \tilde{H}^{\text{high}}, X_i).$$

This gives an individual counterfactual prediction for every person. It answers a narrow question: how much would predicted retirement change if everyone had the same high-ISCED median health level, leaving age, gender, education, income, and migration unchanged?

CF2 is the *distributional counterfactual*. Instead of assigning everyone one health value, it asks what the low- and medium-ISCED samples would look like if their full baseline health distribution resembled that of the high-ISCED group. We implement this with DiNardo–Fortin–Lemieux reweighting (DiNardo et al., 1996). For each source group $s \in \{\text{low ISCED}, \text{medium ISCED}\}$, we pool group s with the high-ISCED target group and estimate a probit for high ISCED using Health PC1, Health PC1 squared, age, and gender. If $\pi_i = \Pr(\text{high ISCED} \mid H_i, H_i^2, \text{age}_i, \text{gender}_i)$, each source-group observation receives a reweighting factor proportional to

$$\omega_i^s \propto \frac{\pi_i}{1 - \pi_i}.$$

The code multiplies this odds ratio by a group-specific constant; this has no effect on the reported CF2 means or densities because the weights are normalised within each source group. The CF2 mean rate is then the weighted mean of the original logit predictions for the source group:

$$\hat{R}_s^{CF2} = \frac{\sum_{i \in s} \omega_i^s \hat{p}_i}{\sum_{i \in s} \omega_i^s}.$$

Thus CF2 does not create a new prediction by changing each person’s covariates. It changes the weights attached to lower-ISCED respondents so that their weighted health distribution moves toward the high-ISCED target. Thus, we interpret the distribution for CF2 in Figure 12 as the weighted distribution of the original individual predicted probabilities, not a set of mechanically edited individual health values.

Figure F.1 shows what the DFL reweighting does mechanically: the reweighted low- and medium-ISCED Health PC1 densities move toward the high-ISCED target distribution.

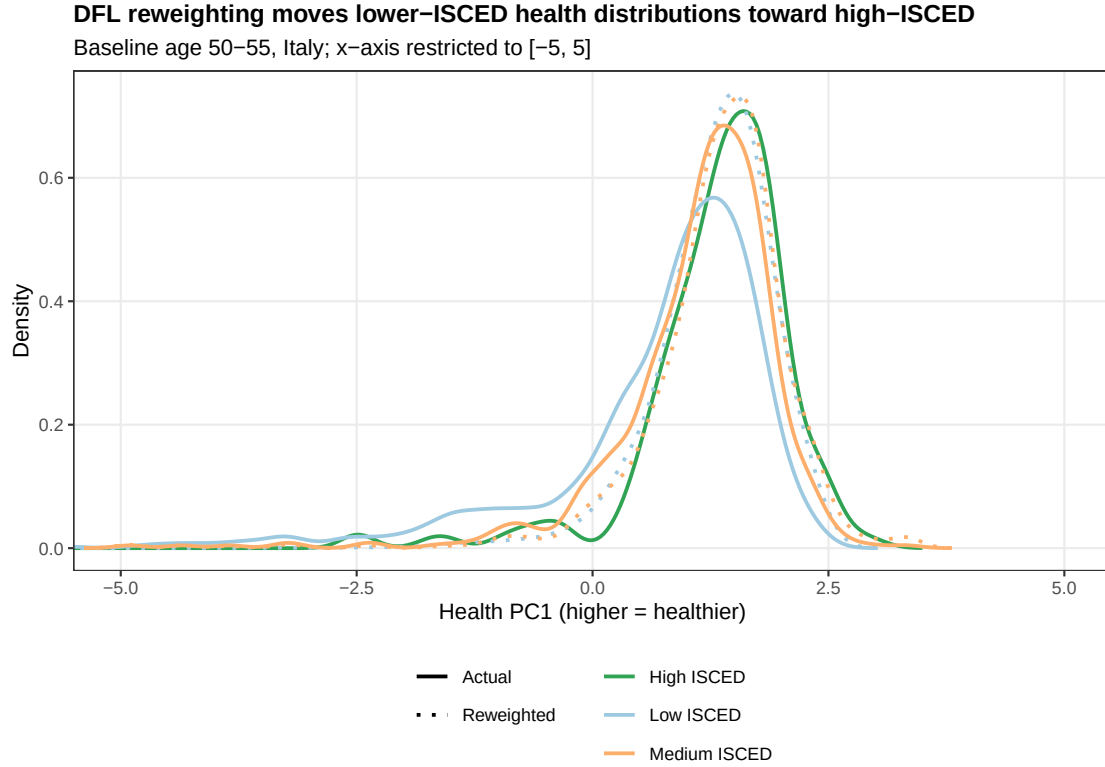


Figure F.1: DFL reweighting shifts the low- and medium-ISCED PC1 distributions toward the high-ISCED target distribution (baseline age 50–55).

F.3 Rockwood frailty item list

The 30 binary deficits used to compute `frailty_rockwood` are: (1) self-reported health fair/poor (`sphus` ≥ 4); (2–3) one or two chronic conditions; (4–5) one or two ADL limitations; (6–7) one or two IADL limitations; (8–9) one or three mobility limitations; (10–12) at least one large-muscle, gross-motor, or fine-motor limitation; (13) `EURO-D` ≥ 4 ; (14–15) BMI outside [18.5, 30) or ≥ 35 ; (16) current smoker; (17–18) immediate and delayed recall ≤ 2 ; (19) orientation ≤ 2 ; (20) numeracy ≤ 2 ; (21) prescribed medicines for hypertension; (22) overnight hospital stay in the past year; (23) activity limited by health; and (24–30) seven available `EURO-D` symptom indicators. We require at least 20 non-missing deficits for a valid index. The index is a deficit-accumulation summary rather than a causal model of frailty; the separate `EURO-D` symptoms give mental health multiple deficit items, so the PCA index is used as the cleaner scalar health measure in the main regressions.

F.4 Concentration indices of health inequality

Figure 7 reports income-related concentration indices for Rockwood frailty. These indices answer a simple distributional question: after ranking respondents from poorest to richest, is frailty spread evenly across the income distribution, or is it concentrated among poorer respondents?

The concentration curve in Figure F.2 gives the intuition. It cumulates the share of total frailty as we move from the bottom to the top of the income ranking. If frailty were unrelated to income, the curve would lie on the 45° line. In the Italian data the curve lies above that line: the bottom 30% of the income distribution accounts for

about 34% of total frailty, and the bottom 50% accounts for about 56%. This is why the concentration indices are negative.

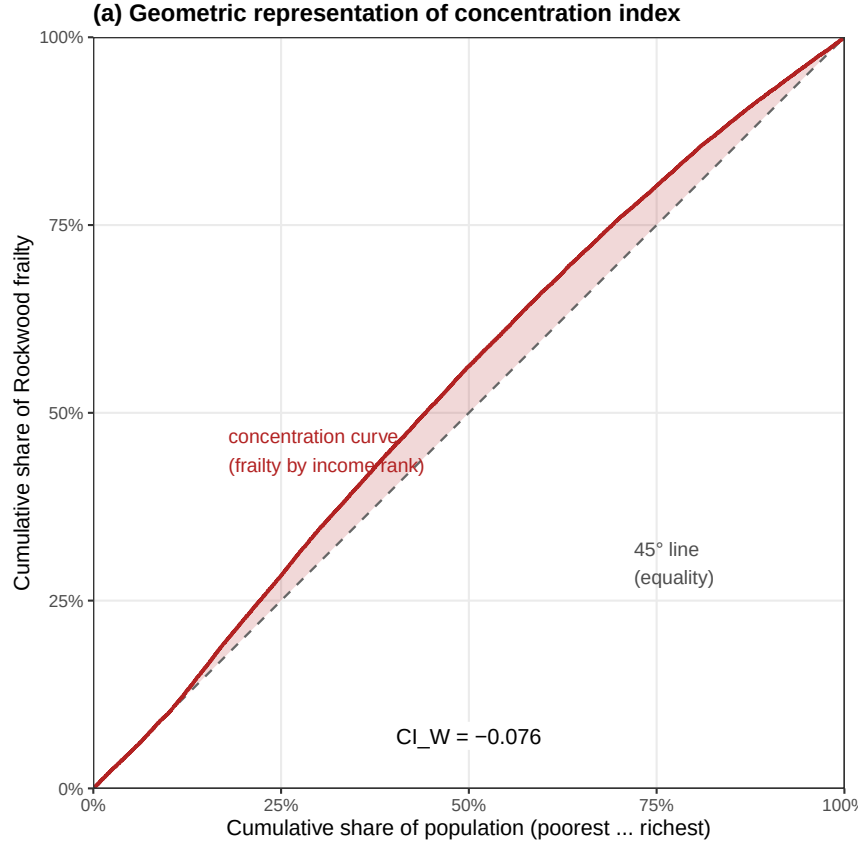


Figure F.2: Concentration curve of Rockwood frailty against income rank in Italy (pooled across waves). Curve above the 45° line indicates frailty is concentrated among the income-poor.

The Wagstaff index is the standard summary of this curve in the health-inequality literature:

$$CI_W = \frac{2}{\bar{y}} \text{cov}(y_i, R_i).$$

Here y_i is frailty and R_i is the respondent's fractional income rank. A value of zero means no income gradient. A negative value means an ill-health variable is concentrated among poorer respondents; a positive value would mean concentration among richer respondents.

We also report the Erreygers index because Rockwood frailty is bounded between zero and one. For bounded outcomes, movements in the Wagstaff index can partly reflect changes in the average level of frailty, not only changes in inequality. The Erreygers correction rescales the same underlying concentration curve in a way that is more comparable across waves:

$$E(y) = \frac{4\bar{y}}{b_{\max} - b_{\min}} \cdot CI_W = \frac{8}{b_{\max} - b_{\min}} \text{cov}(y_i, R_i).$$

In this paper the substantive reading is the same for both measures: frailty is income-related in every wave and is always concentrated among lower-income Italians. The two series are reported together because Wagstaff is the familiar benchmark, while Erreygers is the cleaner over-time comparison for a bounded frailty index.